

Article

A Unified Framework for Detection of ADHD Using EEG Signals and Coherent Models

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Abstract

A behavioral and neuropsychological disorder that develops in young children during their early school years is called attention-deficit hyperactivity disorder (ADHD). When young children are diagnosed with ADHD, they have a tendency not to concentrate on academic and extracurricular activities. Moreover, children affected with ADHD suffer from mood swings, so it becomes quite difficult for them to establish good connections with teachers and friends. In the field of clinical research, deploying Electroencephalography (EEG) signals, a rapid and accurate diagnosis of ADHD is essential so that an effective treatment can be given to the children affected with ADHD. In this work, a unified framework is proposed for the detection of ADHD using EEG signals and some coherent models. The framework initially employs the concept of normalization of EEG signals, followed by the usage of dimensionality reduction techniques such as Local Linear Embedding (LLE), Sammon Mapping (SM) and Locally Linear Coordination (LLC). The dimensionally reduced EEG values are further clustered using four techniques such as spectral clustering, K-means clustering, Fuzzy C-means (FCM) clustering, Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN), and finally, silhouette coefficient analysis is used to analyze the clustering effectiveness. The features are then extracted from the clustered values using an Improved Wavelet Transform (IWT) and then the features are selected with four efficient techniques such as the chi-squared test, Mutual Information (MI), Mahalanobis analysis and Binary Horse Herd Optimization (BHHO) techniques. Finally, the selected values are fed into classifiers for classification with the help of ten traditional machine learning classifiers. The work is tested on a publicly available ADHD dataset and the analysis shows that the best results are obtained when the LLC dimensionality reduction is utilized with FCM clustering and IWT feature extraction, BHHO feature selection, and classified with LGBA classifier reporting a high classification accuracy of 98.12%.



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1. Introduction

Attention-deficit hyperactivity disorder (ADHD) is a behavioral disorder usually diagnosed in childhood [1]. This disorder is characterized by hyperactivity, inattention,

lack of concentration, memory problems, etc. The symptoms of this disorder can start as early as three or four years of age or can even start at the age of ten or eleven years of age [2]. When children are diagnosed with ADHD, it is quite difficult to manage emotions, behavior, thinking patterns, etc., and so managing day-to-day activities becomes so tough for these children [3]. Honing the skills required to survive in this competitive world seems to be very tough for them, and so it results in poor academic growth, which leads to a lot of worry for the parents and family members. Moreover, there are chances of depression accompanied by drug and alcohol abuse, which can also lead to anti-social activities sometimes [4]. ADHD can coexist with other psychiatric conditions like bipolar disorder, schizophrenia, dementia, etc. The exact identification of ADHD is necessary so that efficient modalities can be developed to improve the quality of life of children affected with this disorder [5]. The diagnosis of ADHD is made based on the data provided by the parents of the child diagnosed with ADHD. Sometimes, teachers, relatives and neighbors, too, can help in the process by answering questionnaires that can help the physician to diagnose ADHD. The diagnosis is highly dependent on the skillset and experience of the physician, and sometimes misdiagnosis can also occur, as the identification of ADHD in general is quite a difficult task [6]. To diagnose ADHD, invasive and non-invasive techniques are used so that neural activity can be recorded, as it is highly appreciated with brain functions. The primary non-invasive techniques used for the diagnosis of ADHD are EEG, functional near-infrared spectroscopy (fNIRS), functional magnetic resonance imaging (fMRI) and magnetoencephalography (MEG) [7]. When other techniques are compared, EEG is comparatively better as it is quite inexpensive and non-invasive in nature, and so it is used for tracking the activities of the brain. When ADHD studies in the past were conducted, the EEG recording was done either at the resting state condition or under various cognitive tasks [8]. Computing the linear and nonlinear features of EEG signals and classifying them with machine learning models has been considered by most methods in the past for the successful identification of ADHD. Various techniques are hybrid and employed to diagnose ADHD, as this disorder has a very complex nature and obtaining the necessary diagnostic data from EEG signals is so hectic [9]. Some of the famous works done in the past for the analysis of ADHD are as follows.

A multicenter validation concept of using EEG for identifying ADHD, where the main power and ratio of rhythms ideas were discussed in detail by Synder et al., reported an accuracy of 89% [10]. A decision support algorithm based on Power Spectrum Density (PSD)-based analysis for diagnosing ADHD using EEG was done by Abibullaev and reported an accuracy of 97% [11]. The deficit or hyperactivity disorder versus the normal controls were classified with the help of the quantitative EEG theta/beta ratio concept, and an accuracy of 85% was reported by Ogrim et al. [12]. The non-linear features and neural network were used by Mohammadi et al. for classification of ADHD using EEG, reporting a classification accuracy of 93.65% [13]. The concept of attention continuity, along with Katz and Higuchi' ideas were used for detecting ADHD by Allahverdy et al., reporting a classification accuracy of 96.7% [14]. With the usage of multiple nonlinear dynamical features of EEG signals, the functional brain dynamic analysis of ADHD and control children was conducted by Khoshnoud et al., reporting a classification accuracy of 83.33% [15]. Another framework of classification of ADHD using nonlinear analysis of EEG signals was reported by Boroujeni et al., reporting a classification accuracy of 96.05% [16]. The concept of transfer learning was used to identify ADHD with the help of EEG recordings, and an accuracy of 94.39% was reported by Zhang et al. [17]. The concept of frequency band separation and of a Convolutional Neural Network (CNN) based on continuous mental tasks using EEG to diagnose ADHD was done by Moghaddari et al., reporting an accuracy of 98.48% [18]. The concept of using the directed phase

transfer entropy of EEG signals along with connecting matrices was utilized by Ekhlesi et al. for the classification of children with ADHD, reporting an accuracy of 89.1% [19]. The concept of variational mode decomposition (VMD) and Hilbert transform-based EEG rhythm separation for the detection of ADHD was done by Khare et al., achieving an accuracy of 99.95% [20]. The connectivity patterns of EEG signals were analyzed in detail to discriminate between the linear and nonlinear patterns by Talebi et al., so that ADHD can be diagnosed and detected with a classification accuracy of 99% [21]. An eight-pointed star pattern learning concept for analyzing ADHD using EEG signals was proposed by Tanko et al., reporting an accuracy of 97.19% [22]. The concept of dynamic connectivity patterns with convLSTM and an attention framework was analyzed by Bakhtyari et al., reporting an accuracy of 99.75% [23]. A novel ternary motif pattern-based ADHD detection model using EEG signals was analyzed by Barua, where the concept of Tunable Q-wavelet transforms (TQWT)-based features, along with KNN classifiers, was utilized, reporting a classification accuracy of 95.57% [24]. The concept of non-linear features with tunable Q-wavelet transform and an Artificial Neural Network (ANN) classifier was used by Joy et al. for the detection and classification of ADHD, where a maximum accuracy of 100% was achieved [25].

An explainable and interpretable model using the concepts of Local Interpretable Model-agnostic Explanations (LIMEs), Shapley Additive Explanations (SHAPs) and Partial Dependence Plots (PDPs) was analyzed by Khare and Acharya, reporting an accuracy of 99.81% [26]. The decomposition of EEG signals was done into different sub-bands by means of VMD and robust local model decompositions, and it was fed to deep learning concepts, earning a classification accuracy of over 95% by Esas and Latifoglu [27]. The brain waves were analyzed in detail with the help of a Gaussian Support Vector Machine (SVM)-based model to identify ADHD in children, and a higher accuracy of about 94.2% was obtained by Alim and Imtiaz [28]. The concept of using optimal channels and feature selection technique-based ADHD detection from EEG signals using multiple statistical analysis and conventional machine learning classifiers was done by Maniruzzaman et al., obtaining a high accuracy of 97.53% [29]. A Gabor filter-based statistical feature analysis for the detection of ADHD was proposed by Sathiya et al., reporting a classification accuracy of 96.05% [30]. The EEG feature maps were analyzed along with the Minimum Redundancy Maximum Relevance (mRmR) feature selection technique and deep learning concepts, and a high classification accuracy of about 99.9% was reported by Cura et al. [31]. The concept of continuous wavelet transform (CWT) with gradient-weighted class activation mapping was used by Loh et al. to classify ADHD, and an accuracy of 93.70% was obtained [32]. An EEG-based multi-resolution analysis and feature selection concept was used for ADHD detection and classification by Attallah, obtaining a high accuracy of 99.1% [33]. The concept of Gradient-weighted class activation mapping (Grad-CAM) with a Siamese-based Convolution Neural Network (CNN) was used to analyze EEG so that ADHD can be detected by Latifi et al. [34] and a classification accuracy of 99.17% was obtained. The concept of spectrogram and CNN was used for the diagnosis of ADHD, reporting an accuracy of 97.67% by Abedian et al. [35]. The effect of preprocessing concepts and temporal segmentation on classification accuracy when dealing with ADHD diagnosis with EEG was covered by Ponsoda et al., who reported a high classification accuracy of 86.1% [36]. An entropy difference-based EEG channel selection technique for automated detection of ADHD using Discrete Wavelet Transform (DWT), Empirical Mode Decomposition (EMD), symmetrically weighted local binary pattern (SLBP) and some machine learning classifiers was proposed by Maheshwari et al. and reported a classification accuracy of 99.29% [37]. The concept of Graph Convolution Neural Network (GCN) was utilized for ADHD detection, utilizing multi-domain EEG features, where a high accuracy of 97.29% was achieved by Li et al. [38].

The concept of neuro-discrete cosine transforms independent component analysis with Rhino fish optimization and deep learning was utilized by Alhussen et al., where a high classification accuracy of 98.52% was obtained [39]. Multidimensional EEG characteristics such as Fuzzy entropy, Power Spectral Density (PSD) and Mutual Information (MI) were analyzed and tested on machine learning techniques, and a high classification accuracy of 99.58% was obtained for ADHD detection and classification by Mao et al. [40]. The autoencoder feature mechanism was completed with a ResNet concept-based double augmented attention mechanism method, which yielded a high accuracy of 99.42% by Bansal et al. [41]. In this work, a unified framework is proposed for the detection of ADHD using EEG signals and some coherent models. The overall simplified block diagram of the work is shown in Figure 1 as follows.

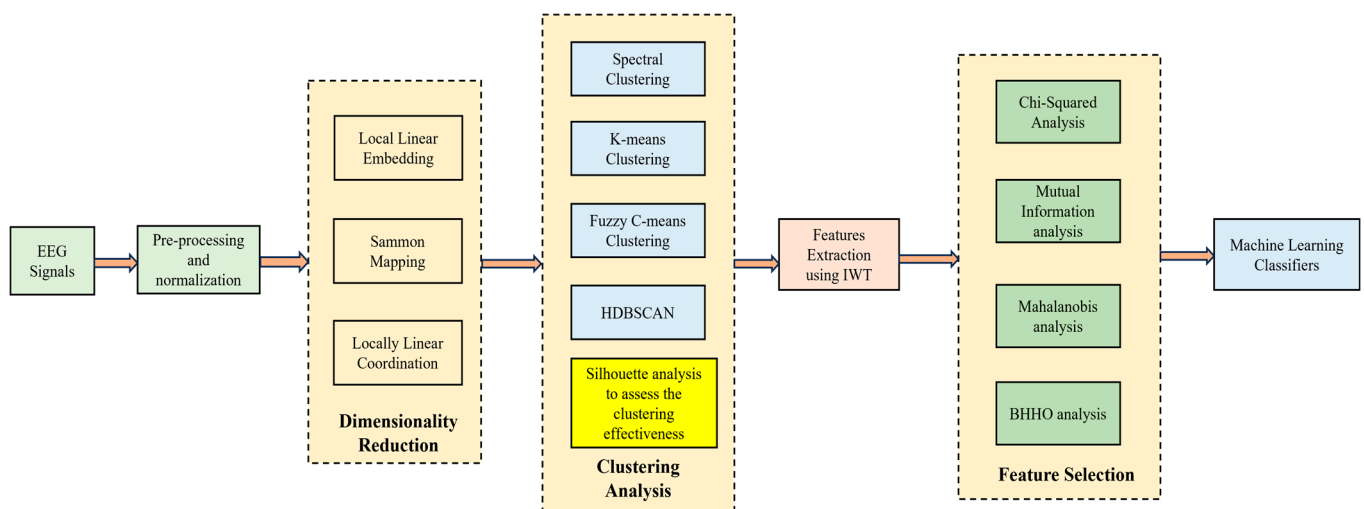


Figure 1. Overall simplified illustration of the work.

- (A) The framework initially employs the concept of basic preprocessing using Independent Component Analysis (ICA), followed by the normalization of EEG signals, and it proceeds by the usage of dimensionality reduction techniques such as Local Linear Embedding (LLE), Sammon Mapping (SM) and Locally Linear Coordination (LLC).
- (B) The dimensionally reduced EEG values are further clustered using four techniques, such as spectral clustering, K-means clustering, Fuzzy C-means clustering, and Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN), and finally, silhouette coefficient analysis is used to analyze the clustering effectiveness.
- (C) The features are then extracted from the clustered values using an Improved Wavelet Transform (IWT), and then the features are selected with four efficient techniques, such as chi-squared test, Mutual Information (MI), Mahalanobis analysis and Binary Horse Herd Optimization (BHHO).
- (D) Finally, the selected values are fed into classifiers for classification with the help of ten traditional machine learning classifiers.

The organization of the paper is as follows. Section 2 discusses the implementation of the proposed coherent model, which includes normalization and dimensionality reduction in input EEG signals, followed by the clustering and feature extraction processes. The different feature selection techniques utilized in this work are discussed in Section 3 and the classification details are elaborated in Section 4. The results are expressed in Section 5, the discussion part is done in Section 6, and the paper is concluded in Section 7.

2. Implementation of the Proposed Coherent Model

Initially, in this section, the normalization, dimensionality-mitigation aspects of input EEG signals, and clustering and feature extraction procedures are discussed in detail.

2.1. Normalization of Input EEG Signals

Once the basic pre-processing is done, the EEG signals are standardized by implementing Z-score normalization [42] in this work, which ensures that the data has a mean of 0 and a standard deviation of 1. The recorded EEG signals can be brought to a common scale with the help of Z-score normalization. Every EEG signal can be centered around zero with the help of Z-score normalization and unit variance can be achieved. The influence of outliers is minimized in this process so that the data transformation can be done to a standard scale. The Z-score normalization can be mathematically expressed as follows:

$$Z = \frac{P - \mu}{\sigma} \quad (1)$$

The actual EEG signal dataset is indicated by P , μ represents the mean and σ represents the standard deviation. Z indicates the normalized value. The central tendency of the signal value is understood well, and so the mean of the EEG signal is computed well, as follows:

$$\mu = \frac{1}{N} \sum_{i=1}^N P_i \quad (2)$$

The value of the individual signals is indicated by P_i and the total sample number is specified by N . With the help of the standard deviation around the mean, the spread of the signal values is computed as follows:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (P_i - \mu)^2} \quad (3)$$

Once these basic computations are done, the estimation of the Z-score normalization is done using Equation (1), so that the output becomes a transformed EEG signal. When the outcomes are obtained, they are focused on zero so that a standard normal distribution is followed, allowing them to be further analyzed.

When the concepts of dimensionality reduction, feature extraction, feature selection, clustering, and machine learning are employed, a good performance can be achieved by the overall model. Complex relationships can be easily visualized by this technique and the overall effectiveness of the methodology is improved. Issues such as redundancy, high sparsity and overfitting problems can be alleviated if this methodology is adopted. Once the normalization of the EEG signal is done, the concept of dimensionality reduction is incorporated before proceeding with clustering, feature extraction, feature selection and classification methodology.

2.2. Dimensionality Reduction in EEG Signals

Now the normalized EEG dataset is specified in a $n \times D$ matrix P comprising n data vector $p_i (i \in \{1, 2, \dots, n\})$ with a certain dimensionality D . An intrinsic dimensionality d is possessed by this dataset and represented as $d \ll D$. The dataset P with dimensionality D is transferred into a new dataset Q with dimensionality d with the help of the dimensionality reduction technique.

(A) LLE as a dimensionality reduction technique:

As far as LLE is concerned, it is quite similar to the Isomap technique. Initially, a graphical indication of the data points is constructed. The local properties of the data are preserved and managed well by the concept of LLE [43]. The non-convex manifolds are embedded successfully with the help of the LLE technique. The high-dimensional data

points are scripted as a linear combination of their corresponding neighbors so that the construction of the local properties of the data is done entirely. The reconstruction weights are retained well by LLE when the representation of the data is in low dimension manifold. Around a data point p_i , the manifold's local properties are expressed well by LLE so that the data point can be expressed as a linear amalgamation of the reconstruction weights w_i with its respective k nearest neighbors p_{ij} . The manifold is assumed to be locally linear so that a hyperplane can be managed well through a data point p_i in the LLE concept. For the data point p_i , the respective reconstruction weights w_i are invariant to rescaling, translation and rotation. The reconstruction weights are preserved well in the context of low dimensionality space by means of assessment of the hyperplane linear mapping. For the manifold, the local geometry is preserved well, and so the data is represented well in a low dimension. The data point p_i is reconstructed well from its neighbors, with the help of reconstruction weights w_i , so that data point q_i is reconstructed well from its neighbors, even if the representation of the data is analyzed in low dimensions. Minimizing the cost function is essential so that the d -dimensional data specification Q can be found out as follows:

$$\phi(Q) = \sum_i \left\| q_i - \sum_{j=1}^k w_{ij} q_{ij} \right\|^2 \text{ subject to } \|q^{(k)}\|^2 = 1 \text{ for } \forall k \quad (4)$$

where the k^{th} column of the solution matrix Q is indicated as $q^{(k)}$. The trivial solution $Q = 0$ is excluded so that the covariance of the column of Q is analyzed well. With respect to the minimum d non-zero eigenvalues, the eigenvectors are computed and the reconstruction weights are analyzed accordingly.

(B) Sammon Mapping as a dimensionality reduction technique:

Sammon mapping tries to concentrate on managing both the small pairwise distances and the large pairwise distances [44]. The classical scaling cost function is adapted well by Sammon mapping, so that the contribution of every pair (i, j) is well-weighted to the cost function. An equal number of weights is assigned to the cost function so that the pairwise distances are retained well. Ultimately, the local structure of the data is retained well, and it is much better than classical scaling. The Sammon cost function is mathematically expressed as follows:

$$\phi(Q) = \frac{1}{\sum_{ij} d_{ij}} \sum_{i \neq j} \frac{(d_{ij} - \|q_i - q_j\|)^2}{d_{ij}} \quad (5)$$

where the pairwise Euclidean distance is indicated as d_{ij} . The main analysis is done between the high-dimensional data points p_i and p_j so that the cost function gradient can be simplified. With the help of a pseudo-Newton technique, the Sammon cost function is minimized.

(C) Locally Linear Coordination (LLC) as a dimensionality reduction technique:

The local linear models are computed with the help of LLC, and then the linear models are globally aligned in a subsequent manner [45]. There are two important steps in this procedure; initially, a local linear model is computed with the aid of an Expectation Maximization (EM) algorithm, and then secondly, the local linear models are aligned so that the low-dimensional data specification is obtained with the help of a variant of a local linear embedding (LLE). A mixture of f factor analyzers are constructed initially using the EM algorithm or, instead of it, a probabilistic Principal Component Analysis (PCA) model could also be constructed. The f data representations d_{ij} are constructed with the help of local linear models and their respective responsibilities r_{ij} are assigned to each data point p_i . The correspondence of every data point p_i to the model is expressed by this

responsibility r_{ij} and the condition $\sum_j r_{ij} = 1$ should be satisfied. These data representations are assigned a weighted responsibility and computed as $u_{ij} = r_{ij}d_{ij}$ and then stored in a block matrix M . Depending on U and on the matrix M , the local models are aligned as $M = (I_n - W)^T(I_n - W)$. The reconstruction weights are specified by W and computed with the aid of Local Linear Embedding (LLE). The identity matrix with a $n \times n$ format is represented by I_n . The generalized eigenproblem is solved by means of the alignment of the local models by LLC for the d smallest nonzero eigenvalues and is represented as:

$$Hv = \lambda Gv \quad (6)$$

The product of $M^T U$ is represented by H and the product of U is represented as G . A matrix K is formed with the help of d eigenvectors v_i so that the linear mapping is defined by means of assigning the weighted data specification U to the low-dimensional data specification Q . By means of calculating $Q = UK$, the representation of the data in low dimensions is found easily.

2.3. Clustering Process

Once the dimensionality reduction is carried out, the concept of clustering is incorporated in the work. Four types of clustering are incorporated, such as Spectral Clustering (SC), K-means clustering, Fuzzy C-means clustering (FCM), and Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN), and silhouette coefficient analysis is then performed to assess the effectiveness of clustering in the experiment.

(A) Spectral Clustering:

Among the data points, a similarity graph is constructed by this type of clustering [46]. The data is grouped into different clusters with the aid of eigenvectors of the Laplacian mode. For dealing with complicated cluster boundaries and non-convex structures, spectral clustering is highly useful.

(B) K-means Clustering:

It is a famous centroid-based algorithm where the data is partitioned into K number of clusters [47]. Within every group of data, the variance is minimized, and so the best performance is always aimed for. When the clusters have similar variance values, K-means clustering works more efficiently.

(C) FCM:

FCM is basically a soft clustering algorithm where the data points are assigned to multiple clusters with a different membership degree [48]. A membership grade or probability is always present and has a value from 0 to 1, showing how much a particular point can belong to a cluster. The mean of all the data points is computed as a centroid and weighted by its membership degree. Until convergence is achieved, the cluster centers and membership values are updated iteratively; this technique is somewhat less sensitive to noise and outliers.

(D) HDBSCAN:

In this method, the clusters of different densities and shapes are identified initially as a density-based algorithm [49]. The knowledge about the total number of clusters is not required to be known in advance, and the density-based algorithm is extremely robust to noise.

(E) Silhouette coefficient analysis to assess the clustering effectiveness:

To assess the clustering effectiveness, the metric utilized is the silhouette coefficient [50]. In the low-dimensional space, clustering tightness within the samples for the similar

categories can be reflected well with the help of this coefficient and is mathematically expressed as follows:

$$s_{il}(i) = \frac{v(i) - u(i)}{\max(u(i)v(i))} \quad (7)$$

where the average distance from data point i to every other data point present in the same cluster is expressed as $u(i)$. The minimum average distance from data point i to every other data point present in a separate cluster is expressed as $v(i)$. For each data point, the average is computed and expressed as the silhouette coefficient as follows:

$$s_{il} = \frac{1}{n} \sum_{i=1}^n s_{il}(i) \quad (8)$$

A good clustering performance is obtained if the silhouette coefficient is higher and proves that it has the potential to be classified correctly.

2.4. Feature Extraction with IWT Process

Once the dimensionality reduction is carried out, the feature extraction process is executed successfully. For the analysis of EEG signals, the wavelet transform is widely used as it is a versatile time-frequency technique dependent on multiple scales [51]. The standard wavelet transform has good focus on decomposing low-frequency components, whereas the least importance is given to high-frequency components, as it is quite insensitive to them. When the wavelet transformation is improvised, the signal decomposition is successful and the local characteristics of the signal can be exhibited well, so that the extraction of features of even the high-frequency signal is done easily [52]. In this concept of IWT, the scale factor is assumed as $H^2(R) = \oplus S_k$ and the division of the Hilbert space $H^2(R)$ is done into an orthogonal sum of subspaces $S_k (k \in \mathbb{Z})$. For the wavelet function $\psi(t)$, the specification of the wavelet subspace is done as S_k . To improve the frequency resolution, the S_k is divided further with the help of the wavelet packets based on the binary system [52]. The representation of the orthogonal space is indicated as follows:

$$w_{k,l}^n(t) = 2^{-\frac{k}{2}} w^n(2^{-k}t - l) \quad (9)$$

The translation factor is represented by l and the first k wavelet is indicated by w_k^n for $n = 0, 1, 2, \dots, 2^k - 1$. The following equations are satisfied by $w_{k,l}^n$ and are represented as follows:

$$w_{k,0}^n = \sum_l f_0(l) w_{k-1,l}^i, \quad n \text{ is even} \quad (10)$$

$$w_{k,0}^n = \sum_l f_1(l) w_{k-1,l}^i, \quad n \text{ is odd} \quad (11)$$

The pair of orthogonal filters is indicated by $f_0(l)$ and $f_1(l)$, $k, l \in \mathbb{Z}$, $n = 1, 2, 3, \dots, 2^k - 1$. The Daubechies wavelet is utilized in this work as it can capture and assess all the variations in the EEG signal efficiently. Moreover, an asymmetrical property is exhibited by the Daubechies wavelet so that the transient changes in the EEG signals can be tracked easily. The essential detection of the EEG signal can be preserved and both the frequency/time domain features can be captured by the Daubechies wavelet. The decomposition levels in the Daubechies wavelet are chosen as eight in this work and a good number of approximations and detailed coefficients are obtained. Thus, even critical features can be extracted in detail from EEG signals with the help of IWT, as even high-frequency components can be decomposed easily, thereby improving the overall characterization of the signal.

3. Feature Selection Techniques Used in the Coherent Model

The extracted features are then chosen with the aid of feature selection techniques. The feature selection techniques considered in this work are chi-squared test, Mutual Information (MI), Mahalanobis analysis and BHHO.

3.1. Chi-Squared Test for Feature Selection

In statistical models, to assess the importance of features, the chi-squared test (χ^2) is highly useful [53]. The chi-squared statistics are computed with respect to a particular class so that the significance of a feature can be assessed well. A null hypothesis H_0 is analyzed initially and then the strong relationship between the feature and its outcome is analyzed in detail. The computation of the chi-squared statistic is done as follows:

$$\chi^2 = \sum_{m=1}^r \sum_{n=1}^c \frac{(Obs_{mn} - Exp_{mn})^2}{Exp_{mn}} \quad (12)$$

Under a null hypothesis H_0 , the observed frequencies (Obs_{mn}) and the expected frequencies (Exp_{mn}) are compared by this test. If the chi-squared statistic is high, then it shows that there is a good correlation between the feature and its outcome. The critical features are traced for the sake of prediction and then the performance of the model is improved by means of eliminating the irrelevant variables. The feature space can be improved by means of implementing the chi-squared test so that the essential components are chosen for classification. Computational efficiency can be improved greatly by eliminating duplicate features so that the generalization of the model is enhanced, and thereby the outcomes can be accurately predicted.

3.2. Mutual Information Analysis for Feature Selection

A filter-based feature selection method is MI. The quantity of information that a random variable possesses about another random variable is computed and measured by MI, thereby tracing and choosing the highly effective features suitable for classification [54]. The value of the MI generally ranges from zero to one. The strong discriminator is identified as the one that has the highest value in the feature subset. The computation of Mutual Information is assumed as follows:

$$MI(P, Q) = \sum_{p \in P} \sum_{q \in Q} \Pr(p, q) \log \frac{\Pr(p, q)}{\Pr(p)\Pr(q)} \quad (13)$$

When the random variables P and Q are evaluated, the representation of the joint probability density function is specified as $\Pr(p, q)$. The individual marginal probability density functions are $\Pr(p)$ for P and $\Pr(q)$ for Q . The joint distribution $\Pr(p, q)$ is compared with the product of the separate marginal distributions $\Pr(p)$ and $\Pr(q)$, and can be represented as $MI(P, Q)$. In the feature selection context using the MI technique, the features are specified by P and the class values are represented by Q .

3.3. Mahalanobis Analysis for Feature Selection

The pertinent attributes are selected with the help of this model so that the model performance is optimized and the dimensionality is mitigated. Assuming a particular point and its related distribution between them is measured with the help of the Mahalanobis distance [55]. The outliers are identified and the variance within the data can be analyzed well. For every feature, the Mahalanobis distance is computed and the degree of divergence can be assessed for each feature. Thus, only the necessary features required for classification are selected and proceed. In this method, the dataset is divided into specific blocks, with several samples contained in every block. Different blocks are utilized in various blocks.

The Mahalanobis distance is computed for every sample in a block, as follows. For all the samples present in the block, the mean vector μ is computed and expressed as follows:

$$\mu = \frac{1}{n} \sum_{i=1}^n p_i \quad (14)$$

where the i^{th} sample in the block is indicated as p_i and the total number of samples in the block is represented as n . For the samples in the block, the covariance matrix is calculated. Among the different features present among the samples, the variance and correlation are captured with the help of covariance matrix C and are represented as follows

$$C = \frac{1}{n-1} \sum_{i=1}^n (p_i - \mu)(p_i - \mu)^T \quad (15)$$

For every sample p_i in the block, the Mahalanobis distance is computed using the mean vector μ and the inverse of the covariance matrix C^{-1} . The representation of the Mahalanobis distance is represented as follows:

$$D(p_i) = \sqrt{(p_i - \mu)^T C^{-1} (p_i - \mu)} \quad (16)$$

For all the samples in the block, the Mahalanobis distance is computed, and so the average distance for the entire block is finally computed. For every block present inside a feature, a divergence measure is analyzed with the help of this step. Within the column, the average distance for all the blocks is obtained, and finally, the overall average distance is also computed, thereby the divergence factor for all the features can be specified with the aid of a single representative measure. In every column, the overall average distance is sorted out in the final step depending on the maximum values obtained. The columns are represented by the features, and only the ones that possess the highest average distances are chosen and proceed for classification. The Mahalanobis distance-based assessment thus proves to be a versatile and resilient technique in the context of feature selection.

3.4. Horse Herd Optimization Algorithm for Feature Selection

This famous algorithm is used in multiple applications and was proposed in [56]. The behavior and attitude of six common horses of various ages are considered in this approach and serve as the primary motive for the formation of this algorithm. The social and individual behavior of horses is emulated by HHO so that the search space can be explored and exploited effectively. To achieve this balance, six main behaviors are included, namely, the Hierarchy mechanism (H_m), Imitation mechanism (I_m), Grazing mechanism (G_m), Roaming mechanism (R_m), Sociability mechanism (S_m) and Defense mechanism (D_m). The following equation is used to model the movement of every horse in each iteration, as follows:

$$Z_q^{t,A_{ge}} = \vec{V}_q^{t,A_{ge}} + Z_q^{(t-1),A_{ge}}, A_{ge} = \mu, \sigma, \omega, \tau \quad (17)$$

The position of the q^{th} horse is indicated by $Z_q^{t,A_{ge}}$ where the age range of the horse is indicated by A_{ge} in the present iteration t and the velocity vector is represented as $\vec{V}_q^{t,A_{ge}}$. For mathematical convenience, the classification of every horse in the entire population is done in any one of the four groups, depending on the age, as follows. τ (0–5 years), ω (5–10 years), σ (10–15 years) and μ (15+ years). Depending on the outcomes, the horses are categorized based on the sorted matrix. The top 10 percent of horses are assigned to group μ , the next 20 percent are assigned to σ , the next 30 percent are assigned to ω and the rest 40% is assigned to τ , respectively. Depending on the age of the horse, the measurement

of the velocity is done. Depending on the age factor, the velocity vector of each horse is computed using Equations (18)–(21) as follows:

$$\vec{V}_q^{t,\mu} = \vec{G}_{m_q}^{t,\mu} + \vec{D}_{m_q}^{t,\mu} \quad (18)$$

$$\vec{V}_q^{t,\sigma} = \vec{G}_{m_q}^{t,\sigma} + \vec{H}_{m_q}^{t,\sigma} + \vec{S}_{m_q}^{t,\sigma} + \vec{D}_{m_q}^{t,\sigma} \quad (19)$$

$$\vec{V}_q^{t,\omega} = \vec{G}_{m_q}^{t,\omega} + \vec{H}_{m_q}^{t,\omega} + \vec{S}_{m_q}^{t,\omega} + \vec{D}_{m_q}^{t,\omega} + \vec{I}_{m_q}^{t,\omega} + \vec{R}_{m_q}^{t,\omega} \quad (20)$$

$$\vec{V}_q^{t,\tau} = \vec{G}_{m_q}^{t,\tau} + \vec{I}_{m_q}^{t,\tau} + \vec{R}_{m_q}^{t,\tau} \quad (21)$$

The important steps of independent intelligence and social mechanisms for horses are explained as follows.

3.4.1. Grazing Mechanism

Basically, horses keep grazing on grass fields incessantly for more than sixteen hours daily with very few breaks in between. The modeling of this grazing behavior is close so that the grazing ability of the horse can be identified well. Depending on the decay factor d_g and with every iteration, there is a linear decrease in this parameter. The modeling of the grazing tendency is expressed as follows:

$$\vec{G}_{m_q}^{t,Age} = g_t(u_b + Rl_b) \left[Z_q^{(t-1)} \right], A_{ge} = \mu, \sigma, \omega, \tau \quad (22)$$

The random number is denoted by R and is set in the range of $[0, 1]$. The lower bound and upper bound of the grazing space is denoted by l_b and u_b , respectively, and the values are designed in this experiment in the range of $[0, 1]$. For every age range, the grazing coefficient and value is set to 2 respectively.

3.4.2. Hierarchy Mechanism

The effect of a skilled leader among the horses during its movement is traced by the hierarchy mechanism of this algorithm. Usually, middle aged horses (σ) and (ω) are considered for this criterion and modeled as follows:

$$\vec{H}_{m_q}^{t,Age} = h_q^{t,Age} \left[Z_{best}^{(t-1)} - Z_q^{(t-1)} \right], A_{ge} = \mu, \sigma, \omega \quad (23)$$

$$h_q^{t,Age} = h_q^{(t-1),Age} \times d_h \quad (24)$$

The influence of the leader on the q^{th} horse is indicated by $\vec{H}_{m_q}^{t,Age}$. The position of the leader is ascertained as $Z_{best}^{(t-1)}$ and the decay factor is represented as d_h which helps to mitigate the overall effect with respect to time.

3.4.3. Imitation Mechanism

The learning of the horse is imitated by an imitation mechanism in the HHO algorithm considering both the good and bad attributes. With the help of motion vector $\vec{I}_q^{t,Age}$, the modeling of the imitation mechanism is done, and this behavior is common among the maturing horses (ω). A horse is directed towards the average position ($\hat{Z}_j$) considering the entire group of horses' population (p) and the initiation mechanism is mathematically expressed as:

$$\vec{I}_q^{t, Age} = i_q^{t, Age} \left[\frac{1}{pQ} \sum_{j=1}^{pQ} \hat{Z}_j^{(t-1)} - Z^{(t-1)} \right], Age = \omega \quad (25)$$

$$i_q^{t, Age} = i_q^{(t-1), Age} \times d_i \quad (26)$$

where d_i represents the decay factor and it helps to mitigate the imitation strength over various cycles so that a good convergence is achieved.

3.4.4. Roaming Mechanism

The natural roaming conduct of horses in search of food is examined by this aspect of the algorithm. For young horses (ω), roaming is more prominent when searching for food and diminishes gradually as they mature. The random exploration is indicated by the motion vector for this roaming aspect of the algorithm $\vec{R}_q^{t, Age}$ and it is affected by the factor $r_q^{t, Age}$ which mitigates the decay factor d_r . For all age groups, the ultimate unified equation is specified as follows:

$$\vec{R}_q^{t, Age} = r_q^{t, Age} p Z^{(t-1)}, Age = w, \tau \quad (27)$$

$$r_q^{t, Age} = r_q^{(t-1), Age} \times d_r \quad (28)$$

3.4.5. Sociability Mechanism

The herd conduct of horses is explained well by this sociability mechanism. A lot of importance is given to the horses to progress towards the average position of the entire group of horses. Within the herd, this conduct of sociability assesses two important factors, diversity and security, so that survival can be easy for all the horses. For horses in the age group of 5 to 15 (σ, ω), a high sociability is prevalent and a motion vector $\vec{S}_q^{t, Age}$ is used to express it successfully. With a particular decay factor d_s , the motion vector is mitigated in an iterative manner and is mathematically expressed as follows:

$$\vec{S}_q^{t, Age} = s_q^{t, Age} \left[\left(\frac{1}{Q} \sum_{j=1}^Q Z_j^{(t-1)} \right) - Z_q^{(t-1)} \right], Age = \sigma, \omega \quad (29)$$

$$s_q^{t, Age} = s_q^{(t-1), Age} \times d_s \quad (30)$$

The entire number of horses are represented by Q and the movement of the horse towards the average position of the herd is assessed by $s_q^{t, Age}$ so that the number of cycles can be gradually reduced.

3.4.6. Defense Mechanism

The protective attitude and conduct of the horses serve as an inspiration to build the defense mechanism of the HHO algorithm. During threats and fights, the escape mechanism is incorporated by horses and this mechanism is symbolized by the defense factor ' d ', known for defense, indicating the protective mechanism of the horse. To avoid multiple dangers and to secure resources like water and food, horses must fight often and it is modeled in this algorithm. The worst-performing horses are termed initially and depending on the positions, the escape vector is adjusted in the population. Over time, the updation of factor ' d ' is done so that w_d is decreased in every cycle as follows.

$$\vec{D}_q^{t, Age} = -d_q^{t, Age} \left[\left(\frac{1}{mQ} \sum_{j=1}^{mQ} \hat{Z}_j^{(t-1)} \right) - Z_q^{(t-1)} \right], Age = \mu, \sigma, \omega \quad (31)$$

$$d_q^{t, Age} = d_q^{(t-1), Age} \times w_d \quad (32)$$

The difference between the previous position of the worst horse and the current position of the worst horse is used to affect the escape vector factor, incorporating the above 2 equations.

3.5. Implementation of Binary HHO Algorithm for Feature Selection

The herding behavior of horses thus leads to the development of HHO. In this work, an improvised version of HHO was utilized, known as binary HHO [57]. The feature subset is evaluated in depth and then the relevant features are chosen. Generally, various transfer functions, such as U-shaped, V-shaped and S-shaped, are considered by BHHO so that continuous search spaces can be handled well. Fitness values are scaled approximately by the approach so that the exploration and exploitation capabilities can be manipulated and managed well. The feature selection problem cannot be resolved directly by HHO, as the final solution is generally comprising continuous values. Therefore, to convert the continuous value to binary values such as zero or one, an efficient transfer function is utilized. The transfer function is used to assess the rate of change in the decision variables in the range of 0 to 1. The optimization process generates continuous values and the values are converted to binary values with the help of the transfer function. In the search space, to improve the quality and diversity of the solution, various transfer functions such as V-shaped, S-shaped and U-shaped functions are utilized. In this work, only the S-shaped transfer function is utilized as it provides a good transition for feature selection purposes and can achieve a good exploration/exploitation capacity in the optimization process. The sensitivity of feature selection is controlled by the scaling factors and steepness factors so that the changes in the optimization values can be managed well. To obtain better feature subsets and to get high accuracy, the S-shaped transfer function aids greatly, thereby achieving a good balance between the exploration and exploitation aspects of the algorithm. Thus, a continuous search space can be converted to a binary one with the help of an S-shaped transfer function. For BHHO, the binary solution space is indicated by dimension matrix $n \times N$, where the population size is represented by n and the feature number is represented by N . For mapping processes, the S-shaped transfer function is best as it can give an output in the range of $[0, 1]$, which makes element updates simpler in the process. Some of the S-shaped transfer functions mentioned as ($S_1 - S_4$) are expressed as follows:

$$S_1 = \frac{1}{1 + e^{-2z}} \quad (33)$$

$$S_2 = \frac{1}{1 + e^{-z}} \quad (34)$$

$$S_3 = \frac{1}{1 + e^{-\frac{z}{2}}} \quad (35)$$

$$S_4 = \frac{1}{1 + e^{-\frac{z}{3}}} \quad (36)$$

Across the iterations, consistent scaling can be achieved when using the S-shaped transfer function. A predictable transition can be maintained so that feature selection can be done effectively for high-dimensional data. For every individual z , the continuous search can be processed into a binary one with the help of the following equation and the z_i can be computed for the S-shaped transfer function as follows:

$$z_i^d(t+1)_{bin} = \begin{cases} 1 & rand < TF(z_i^d(t+1)) \\ 0 & rand \geq TF(z_i^d(t+1)) \end{cases} \quad (37)$$

where the d^{th} dimension of every individual z_i for the current iteration $(t + 1)$ is represented as $z_i^d(t + 1)_{bin}$. The *rand* is used to assess the random numbers mentioned within the range of $[0, 1]$. The probability that a particular transfer function would be implemented for every continuous aspect of the d^{th} agent is indicated by transfer function $(z_i^d(t + 1))$. The transfer function used in this work is S-shaped and so the dimension of the original individual is assigned to 1. The process is assessed as follows.

(a) Initialization Phase:

For a swarm of Q horses which are nothing but the search agents, the initialization phase is assigned in a random manner in this population-based technique. The search for individuals is done in a wide manner in this process, but when the continuous search space domain is concerned, a narrow search process is engaged. The upper bounds and lower bounds are assigned to every potential solution in the range of $[-1, 1]$. A possible solution is signified by each search agent based on the feature selection technique with a dimension D_{im} , so the total number of features in every dataset can be made equal. The prediction error rate can be minimized by means of considering the minimum feature subsets so that the feature selection issue can be characterized well for prediction purposes. Therefore, the important values (“ones”) are recognized and the least significant values (“zeroes”) are rejected in this study. The initial position of every horse is discretized within the swarm and the value is assigned as 0 or 1; thereby, the conversion to binary values is done successfully.

(b) Computing the Fitness Function:

To compute the performance of the optimization algorithm, the fitness function is used. The prediction error rate is used to largely influence the fitness function. For an effective feature selection methodology, there should be a smaller number of features supported by a lower error rate. The following equation is used to compute the fitness value of the considered feature selection mechanism.

$$F_{fitness} = w_1 error(D_{im}) + w_2 \frac{\text{selected feature number}}{\text{entire set of features}} \quad (38)$$

To manage the prediction error rate and to analyze the total number of chosen characteristics in the entire population with size n , the two important variable values are assigned as $w_1 \in [0, 1]$ and $w_2 = 1 - w_1$. For an individual feature subset, the computation of the fitness is done considering two primary factors: the number of selected features and the classification accuracy. The model performance can be balanced well with this design so that complexity can be avoided in the overall model. The chosen feature subset is implemented by the algorithm to a classifier that can be trained on a dataset. The performance of a classifier on a test set or validation set gives an accuracy score. This core score helps to assess the predictive quality of the feature subset and so it serves as a vital indicator in managing the performance of the classifier. The entire number of features are evaluated in the subset by this algorithm so that the model complexity can be mitigated. The computation of the overall fitness of an individual is done considering the sum of the feature count and accuracy. Here, the significance of every objective is reflected by the weights (w_1) and (w_2) assigned to them. The optimization technique is guided by the weighted fitness values and so the best-performing feature subsets are chosen in the process. The effectiveness of the chosen feature subset is analyzed by this fitness computation method and it is done by means of balancing concepts like dimensionality reduction, feature selection and classification accuracy. With a 10-fold cross-validation, the division of the dataset is done and the training set is utilized to train the classifiers on the selected features. The predictive performance of the model is validated with the help of a testing set so that good generalizability is achieved. The ratio of chosen features to the entire set of features, along

with the classification accuracy from the testing test, is hybridized by the fitness function, and so overfitting is minimized. Also, this technique helps to assess the chosen features' relevance for effective implementation across multiple datasets.

(c) Updation of Position:

In the BHHO algorithm, there is an iterative updation of the position of the horses so that continuous values are allowed in the position vectors across multiple iterations. Depending on prior information, the updates can be enabled successfully. The discretization of these continuous values is done after every update so that the fitness of the binary solution is evaluated. Depending on the prediction error rate, the assessment of fitness is done and then the BHHO algorithm can identify the selected features easily. The procedure of BHHO is explored in Algorithm 1 as follows:

Algorithm 1. Implementation of BHHO

Assign the parameters:

Population size (n), number of features, maximum number of iterations max_iter

Start population Z as $z, i = 1, \dots, n$

$t \leftarrow 0$ is defined

while $t < \text{max_iter}$

For each horse

Utilize an S-shaped transfer function so that horses can be transformed into binary space

Use a simple Naïve Bayesian Classifier (NBC) to assess and evaluate every horse in the population

Analyze the fitness of the horses in the entire population

Determine the age of the horses; $\mu, \sigma, \omega, \tau$

Update and improve the velocity

Update and improve the horse position

Update every horse position using Levy flight

End for

$z_{best} \leftarrow \text{Best Solution}$

Implement crossover operation on z_{best}

Determine a novel candidate solution

Evaluate z_{New} using NBC classifier

Measure fitness value of z_{New}

If $f(z_{New}) < f(z_{Best})$

$z_{Best} \leftarrow z_{New}$

End if

$t \leftarrow t + 1$

End while

Return z_{best}

4. Classification Process Used in the Coherent Model

The selected features are fed to classifiers to assess the classification accuracy. The classifiers used in the work are Random Forest (RF), Gaussian Naïve Bayes (GNB), Logistic Regression (LR), Extreme Gradient Boosting (XGB), Naïve Bayesian Classifier (NBC), K Nearest Neighbor (KNN), Decision Trees (DT), Support Vector Machine (SVM) and Light Gradient Boosting Approach (LGBA) [58].

4.1. RF Classifier

It is quite a famous ensemble learning technique where multiple decision trees are created so that randomness can be added to it to mitigate the predictor correlation. The samples are bootstrapped in this classifier so that at every split the variables are selected, and this gives a high stability to the entire process. With the aid of majority voting, an accurate prediction can be made for a variety of classification tasks when using an RF classifier.

4.2. GNB Classifier

In this classifier, a lot of predictors are present, and the principle of independence is applied by this method with the help of Bayes theorem. A Gaussian distribution is utilized so that the continuous attributes are hypothesized in this classifier, so that overfitting problems are usually avoided, and it can be applied for multiple parallelized classification tasks.

4.3. LR Classifier

The likelihood of an input is predicted well by the concept of LR, so that the input can belong to any one of the two categories. LR works efficiently with datasets comprising more than one independent variable. A logistic function is used by this classifier so that the input data is converted into a specific probability outcome, thereby the dataset with a linear relationship can be managed well.

4.4. XGB Classifier

In this famous ensemble learning technique, decision trees are built in a sequential manner so that the previous mistakes are fixed and corrected. It can handle large datasets efficiently and can process with good speed, and so it is used generally for both regression and classification tasks.

4.5. NBC

Depending on Bayesian decision theory, NBC was developed, and the class of every input can be predicted easily by means of computing the posterior probability of every class. Once the posterior probability of every class is computed, then the class with the highest probability is chosen. Features like probability density function, prior probability and conditional density are often considered when using NBC. To assess and calculate computational efficiency, the Gaussian distribution is used in NBC.

4.6. KNN Classifier

A famous supervised learning technique is the KNN algorithm and it is used for multiple applications in the context of machine learning. Data distribution is not assumed in this process as it is non-parametric in nature. The training data is often stored by KNN and so it is referred to as lazy learning, and, depending on the frequent label points, the classification of the new points is done amongst the KNN. The KNN is quite simple to implement and the proximity to neighboring points can be evaluated well so that the capability of data classification can be emphasized well by this concept.

4.7. DT Classifier

This supervised learning algorithm ensures a streamlined process, so that good accuracy can be obtained. A root node is the essence of a decision tree, and with the aid of root nodes, feature values are split sequentially until it reaches the leaf nodes, which would specify the prediction of the final class. The independent splits of the feature space are represented by the subtrees in this classifier.

4.8. SVM Classifier

A fundamental method used in most classification and regression tasks is SVM [59]. In this technique, a decision boundary or hyperplane is created so that the datasets can be partitioned into various groups. When the objects are handled from diverse categories, this distinction is necessary. Kernels are utilized in SVM so that items of different classes are differentiated easily in SVM. The data can be mapped easily into a higher-dimensional space with the aid of Kernel techniques so that the hyperplane can separate the classes easily. A decision surface is thus created by SVM so that the groups can be distinguished clearly and prove their efficacy and versatility in managing non-linear classification issues. In this work, only the polynomial kernel and Radial Basis Function (RBF) kernel are utilized.

4.9. LGBA Classifier

An effective mechanism of the gradient boosting algorithm is LGBA and it serves as an efficient classification model [60]. Multiple weak learners are hybridized into a strong classifier with the help of this LGBA algorithm. This method is a famous tree-based technique and it incorporates the concept of gradient boosting analysis, utilized to deal with datasets which are massive. In every iteration, higher weights are assigned to the samples and in this LGBA model, it is hard to classify instances in the minority class to a great extent. The class imbalance is counteracted in an inherent manner by this technique as it utilizes an adaptive weighting mechanism. The underrepresented samples are made sensitive to this model so that even in the minority classes, the classification performance is largely enhanced. The important steps in the LGBA approach are as follows:

The weak learner is initialized using the following equations:

$$f_0(g) = \operatorname{argmin}_j \sum_{j=1}^n l(h_j, c) \quad (39)$$

$$l(h_i, c) = l(h, f(g)) = (h - f(g))^2 \quad (40)$$

where the basis function of the weak learner is represented as $f_0(g)$. The loss function is specified by l and the entire sample amount is indicated as n in the above equation. In a continuous manner, the weak learners are estimated ' T ' number of times.

- (a) To assess the negative gradient of a loss function, the following equation is utilized for a model $g_i \in G (\forall j = 1, 2, \dots, n)$ and is represented as follows:

$$r_{tj} = - \left[\frac{\partial l(h_i, f(g_i))}{\partial f(g_i)} \right]_{f(g)=f_{t-1}g} \quad (41)$$

where the negative gradient's loss function is indicated as r_{tj} .

- (b) The rest of the r_{tj} result is considered an efficient model of a normal and original value.
 (c) For the $\{(g_i, r_{t1}) \dots (g_n, r_{tn})\}$, a regression tree is fit and then a new regression tree $f_t(g)$ is constructed.
 (d) The optimal fit value is assigned as $k = 1, 2, \dots, K$ so that the leaf regions can be assessed well.
 (e) The overall loss function must be minimized and so s_{tk} is utilized as a linear search and a leaf node can be forecasted well as follows:

$$s_{tk} = \operatorname{argmin}_s \sum_{g_j \in R_{tk}} l(h_j, f_{t-1}(g_i + c)), j = 1, 2, \dots, T \quad (42)$$

- (f) With the help of the following equation, the novel robust learning is updated as follows:

$$f_t(g) = f_{t-1}(g) + \sum_{j=1}^K s_{tk} I(g \in R_{tk}) \quad (43)$$

where the existing work learner is indicated as $f_t(g)$.

- (g) The pre-weak learner is indicated as $f_{t-1}(g)$ and the indicator function is specified as I and is represented as follows:

$$I = \begin{cases} 1, g \in R_{tk} \\ 0, g \notin R_{tk} \end{cases} \quad (44)$$

- (h) The final decision tree is determined using the following equation as follows:

$$F(g) = \sum_{t=1}^T \sum_{k=1}^K s_{tk} I(g \in R_{tk}) \quad (45)$$

The significance of an attribute is assessed by the parameter called Gain values or *Gini* and is represented as:

$$Gini(w) = \sum_{l=1}^L w_l(1 - w_l) = 1 - \sum_{k=1}^L w_l^2 \quad (46)$$

where the number of labels is detected by L and the weight of that label is assigned as w_k .

5. Results

The proposed framework was analyzed on a publicly available dataset, which can be obtained from the Dataport of the IEEE [61,62]. The dataset has details of a total of 121 kids, including both girls and boys, in the age group of 7–12. The total number of healthy controls (HC) is categorized into 60 and the total number of kids diagnosed with ADHD is categorized into 61. With the aid of an experienced psychiatrist, the ADHD diagnosis of the kids was performed. The kids belonging to the control category had no history of any neurological or psychiatric disorders. With the help of a 10–20 standard and employing a 19-channel system, the recording of the EEG was done at a sampling frequency of 128 Hz. When the recording of the EEG was done, the reference electrodes were nothing but the electrodes A_1 and A_2 were placed near the earlobes. When the EEG data was recorded, a lot of significance was given to the visual attention tasks. The kids were given images with cartoon characters and letters and they were requested to count the total number of characters in every image. The sizing of the pictures was done appropriately so that the kids can count them very easily. During the EEG recording, a continuous stimulus was maintained, and so the display of every image was done without any disruption. Based on the response speed of the child, the cognitive visual task was managed accordingly when the EEG recordings were carried out. The work was implemented on MATLAB R2022a software installed on a system with an i5 processor, 32 GB main memory and Windows 11 operating system. A 10-fold cross validation method was utilized in this experiment. The classification accuracy was the main metric analyzed in this work and it is computed in terms of True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) and is represented as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (47)$$

When the spectral clustering concept was analyzed, the number of clusters was set as 10 in the work. The eigenmap heuristic is used to assess the total number of clusters, which helps to select the number of clusters. For the K-means clustering concept used, the number of clusters was assigned as 10 in the experiment. As far as the FCM algorithm is concerned, the degree of fuzziness is controlled by the fuzziness exponent and is set to 2 in this work. When the HDBScan algorithm is examined, the minimum cluster size is set to 5, and the maximum cluster size is set to 50 in the experiment. The maximum number of iterations was set as 100 in the experiment for all the clustering techniques. When the Silhouette analysis was done for the clustering algorithms in the experiment, the average

silhouette coefficient ranged from 0.5 to 1, showing a strong, reasonable and well-separated structure, thereby it can be proceeded for the feature extraction and selection stage. As far as the BHHO algorithm is concerned, the population size is set as 100, the gait value is set as 1.5 and the number of iterations is set as 300 in the experiment. The transfer function utilized is an S-shaped transfer function for the BHHO concept. For the RF classifier, the number of trees or estimators is set as 50, as increasing the total number of trees enhances the performance but increases the computational time, and so it is restricted to 50 in the experiment. The number of features sampled and the depth of trees are set at 4 and 1, respectively. As far as the GNB classifier is concerned, the classifier is mostly analytical in nature, and so the estimation of the mean and standard deviation for every feature in the class is done directly from the training data with the aid of maximum likelihood estimation. There is always a naïve assumption in this classification model that the features are conditionally independent of each other. When the LR classifier is concerned, the inverse of the regularization strength is set as 0.1, the regularization techniques chosen are L1 and the maximum number of iterations is set at 100 in the experiment. For the XGBoost classifier, the learning rate is set to 0.2, the maximum depth of a tree is set to 2, the number of estimators is set to 100 and the fraction of data utilized for each tree is set to 0.6 in the experiment. For the NBC classifier, a kernel density estimation and multinomial assignment is set when the classification is performed. For the KNN classifier, the K specifies the number of nearest neighbors utilized to classify a particular point and the value of it is arranged as 10 in the experiment. The distance metric chosen is Euclidean and the distance weight fraction is assigned based on the input feature values. As far as DT is concerned, the maximum number of splits or depth is set as 1, the split criterion is set as 4 and the minimum leaf size is set as 1, respectively. For the polynomial and RBF kernel SVM, the gamma is set to 2, and the value of the regularization parameter is set to 10 in the experiment. For the LGBA classifier, the number of estimators is set as 100, the max_depth value is set as 1, and the learning rate is set as 0.2. The number of leaves is set as 40 to control the complexity of the tree model. The minimal number of data points in a leaf is set as 10, so that it helps to reduce the overfitting in the LGBA classifier, and the L1/L2 regularization term is utilized here so that the complex models can be penalized efficiently.

Upon analysis of Table 1, it is evident that a high classification of 89.93% is obtained when the Chi-square feature selection scheme is utilized with the proposed set of LLE dimensionality reduction, FCM clustering, feature extraction and LGBA classifier. When Table 2 is analyzed, it is observed that a high classification accuracy of 89.98% is obtained when the MI feature selection scheme is utilized with the proposed set of LLE dimensionality reduction, HDBSCAN clustering, feature extraction and LGBA classifier. Upon analysis of Table 3, it is evident that a high classification of 92.71% is obtained when the Mahalanobis-based feature selection scheme is utilized with the proposed set of LLE dimensionality reduction, SC based analysis, feature extraction and LGBA classifier. When Table 4 is analyzed, it is observed that a high classification accuracy of 95.77% is obtained when the BHHO feature selection scheme is utilized with the proposed set of LLE dimensionality reduction, FCM clustering, feature extraction and LGBA classifier.

Upon analysis of Table 5, it is evident that a high classification of 91.67% is obtained when the Chi-square feature selection scheme is utilized with the proposed set of SM dimensionality reduction, FCM clustering, feature extraction and SVM-RBF classifier. When Table 6 is analyzed, it is observed that a high classification accuracy of 92.78% is obtained when the MI feature selection scheme is utilized with the proposed set of SM dimensionality reduction, FCM clustering, feature extraction and LGBA classifier. Upon analysis of Table 7, it is evident that a high classification of 94.68% is obtained when the Mahalanobis-based feature selection scheme is utilized with the proposed set of SM dimensionality reduction,

K-means clustering, feature extraction and LGBA classifier. When Table 8 is analyzed, it is observed that a high classification accuracy of 96.88% is obtained when the BHHO feature selection scheme is utilized with the proposed set of SM dimensionality reduction, FCM clustering, feature extraction and LGBA classifier.

Table 1. Accuracy analysis of the Chi-square feature selection scheme with the proposed set of LLE dimensionality reduction, clustering, feature extraction and classification models.

	RF	GNB	LR	XGB	NBC	KNN	DT	SVM (Poly)	SVM (RBF)	LGBA
LLE + SC	86.34	84.32	83.46	85.56	84.43	82.23	81.02	86.23	87.21	89.01
LLE + Kmeans	87.35	83.26	85.67	84.45	83.46	81.02	81.23	87.23	86.98	88.26
LLE + FCM	88.12	81.48	82.36	87.53	86.01	80.21	82.01	85.99	86.99	89.93
LLE + HDBSCAN	85.36	83.46	84.12	86.65	83.04	82.21	80.29	87.01	87.01	88.99

Table 2. Accuracy analysis of the MI feature selection scheme with the proposed set of LLE dimensionality reduction, clustering, feature extraction and classification models.

	RF	GNB	LR	XGB	NBC	KNN	DT	SVM (Poly)	SVM (RBF)	LGBA
LLE + SC	84.21	85.98	84.09	86.11	83.11	83.09	82.89	84.22	86.09	88.21
LLE + Kmeans	85.34	84.67	83.98	83.23	82.23	80.99	83.98	85.36	85.99	89.33
LLE + FCM	86.56	82.80	84.87	85.47	85.76	82.82	84.22	88.76	87.81	89.58
LLE + HDBSCAN	84.79	81.91	82.22	84.89	86.59	83.33	82.36	86.54	86.35	89.98

Table 3. Accuracy analysis of the Mahalanobis based feature selection scheme with the proposed set of LLE dimensionality reduction, clustering, feature extraction and classification models.

	RF	GNB	LR	XGB	NBC	KNN	DT	SVM (Poly)	SVM (RBF)	LGBA
LLE + SC	88.23	86.09	89.21	90.98	85.12	86.98	85.09	90.09	91.22	92.71
LLE + Kmeans	89.55	87.98	87.23	89.87	87.36	84.89	87.98	91.98	90.24	91.34
LLE + FCM	89.68	88.80	87.46	87.65	84.78	85.77	88.77	90.11	91.57	92.66
LLE + HDBSCAN	88.98	89.21	85.78	87.22	87.98	82.61	89.90	91.23	90.87	90.79

Table 4. Accuracy analysis of the BHHO-based feature selection scheme with the proposed set of LLE dimensionality reduction, clustering, feature extraction and classification models.

	RF	GNB	LR	XGB	NBC	KNN	DT	SVM (Poly)	SVM (RBF)	LGBA
LLE + SC	92.21	89.09	91.21	92.09	88.21	87.09	84.21	92.55	91.21	92.09
LLE + Kmeans	91.56	89.89	90.22	92.89	89.22	85.99	85.22	93.78	93.22	94.89
LLE + FCM	90.79	90.23	92.47	89.66	86.37	86.81	86.35	93.98	94.37	95.77
LLE + HDBSCAN	90.88	90.55	90.89	90.78	88.76	88.23	88.77	94.13	94.89	94.61

Table 5. Accuracy analysis of the Chi-square feature selection scheme with the proposed set of SM dimensionality reduction, clustering, feature extraction and classification models.

	RF	GNB	LR	XGB	NBC	KNN	DT	SVM (Poly)	SVM (RBF)	LGBA
SM + SC	88.11	86.23	82.09	84.87	86.21	83.09	83.11	85.21	88.22	90.11
SM + Kmeans	89.21	86.34	84.98	85.79	87.33	84.88	84.23	84.45	90.24	91.23
SM + FCM	86.35	84.59	84.45	85.82	88.67	82.93	84.67	86.78	91.67	90.78
SM + HDBSCAN	83.68	83.65	85.67	86.17	86.89	84.21	82.98	88.98	89.98	89.98

Table 6. Accuracy analysis of the MI feature selection scheme with the proposed set of SM dimensionality reduction, clustering, feature extraction and classification models.

	RF	GNB	LR	XGB	NBC	KNN	DT	SVM (Poly)	SVM (RBF)	LGBA
SM + SC	90.21	88.09	83.12	86.22	84.77	85.09	86.70	87.09	89.21	91.22
SM + Kmeans	91.55	89.89	85.33	88.34	88.76	82.90	83.32	83.77	89.33	92.34
SM + FCM	92.68	88.77	87.45	83.55	84.54	87.88	87.11	88.89	89.47	92.78
SM + HDBSCAN	88.98	89.12	82.89	83.67	88.38	82.21	84.39	87.21	88.68	88.98

Table 7. Accuracy analysis of the Mahalanobis feature selection scheme with the proposed set of SM dimensionality reduction, clustering, feature extraction and classification models.

	RF	GNB	LR	XGB	NBC	KNN	DT	SVM (Poly)	SVM (RBF)	LGBA
SM + SC	91.21	88.09	91.12	92.09	89.21	89.34	86.09	91.11	93.11	93.24
SM + Kmeans	92.33	89.89	89.33	92.99	91.23	87.56	89.98	92.12	92.25	94.68
SM + FCM	93.46	90.77	92.46	91.87	89.44	89.88	90.88	89.56	93.66	90.76
SM + HDBSCAN	89.78	89.61	90.78	89.32	91.56	90.98	89.71	92.89	91.78	90.84

Table 8. Accuracy analysis of the BHHO feature selection scheme with the proposed set of SM dimensionality reduction, clustering, feature extraction and classification models.

	RF	GNB	LR	XGB	NBC	KNN	DT	SVM (Poly)	SVM (RBF)	LGBA
SM + SC	94.23	92.09	92.21	93.87	90.12	89.08	89.21	93.87	93.66	94.65
SM + Kmeans	93.44	91.08	91.23	91.76	91.33	90.33	90.22	94.77	94.56	96.67
SM + FCM	92.56	92.77	93.45	90.23	90.67	90.46	91.36	92.98	95.78	96.88
SM + HDBSCAN	91.78	91.89	92.67	92.66	92.84	91.71	92.86	95.25	94.34	95.73

Upon analysis of Table 9, it is evident that a high classification of 96.44% is obtained when the Chi-square feature selection scheme is utilized with the proposed set of LLC dimensionality reduction, K-means clustering, feature extraction and LGBA classifier. When Table 10 is analyzed, it is observed that a high classification accuracy of 96.92% is obtained when the MI feature selection scheme is utilized with the proposed set of LLC dimensionality reduction, SC-based analysis, feature extraction and SVM-RBF classifier. Upon analysis of Table 11, it is evident that a high classification of 97.13% is obtained when the Mahalanobis-based feature selection scheme is utilized with the proposed set of LLC dimensionality reduction, K-means clustering, feature extraction and LGBA classifier. When Table 12 is analyzed, it is observed that a high classification accuracy of 98.12% is obtained when the BHHO feature selection scheme is utilized with the proposed set of LLC dimensionality reduction, FCM clustering, feature extraction and LGBA classifier.

Table 9. Accuracy analysis of the Chi-square feature selection scheme with the proposed set of LLC dimensionality reduction, clustering, feature extraction and classification models.

	RF	GNB	LR	XGB	NBC	KNN	DT	SVM (Poly)	SVM (RBF)	LGBA
LLC + SC	91.34	90.11	92.45	93.34	95.55	93.09	92.12	95.11	96.09	96.23
LLC + Kmeans	92.55	92.21	95.67	94.67	96.68	91.91	93.67	94.27	95.98	96.44
LLC + FCM	93.67	94.34	93.88	94.89	94.98	93.22	93.89	94.89	94.23	95.57
LLC + HDBSCAN	91.89	93.57	95.98	95.85	95.10	92.16	91.23	92.34	93.65	93.98

Table 10. Accuracy analysis of the MI feature selection scheme with the proposed set of LLC dimensionality reduction, clustering, feature extraction and classification models.

	RF	GNB	LR	XGB	NBC	KNN	DT	SVM (Poly)	SVM (RBF)	LGBA
LLC + SC	92.21	92.09	93.23	96.09	94.21	95.33	95.09	95.55	96.92	96.09
LLC + Kmeans	93.33	93.99	95.44	97.99	95.22	93.45	94.88	94.61	95.24	95.88
LLC + FCM	94.45	93.83	96.56	94.01	94.35	96.76	95.72	94.78	95.76	96.82
LLC + HDBSCAN	92.78	91.46	93.89	95.22	93.87	93.35	93.13	91.89	94.88	90.35

Table 11. Accuracy analysis of the Mahalanobis feature selection scheme with the proposed set of LLC dimensionality reduction, clustering, feature extraction and classification models.

	RF	GNB	LR	XGB	NBC	KNN	DT	SVM (Poly)	SVM (RBF)	LGBA
LLC + SC	94.21	93.09	93.21	93.09	90.11	90.21	93.02	94.21	96.21	96.99
LLC + Kmeans	94.34	92.88	94.67	91.88	92.25	90.35	94.33	95.33	96.35	97.13
LLC + FCM	93.66	91.93	90.23	90.94	90.87	91.66	92.47	93.46	96.57	97.02
LLC + HDBSCAN	92.78	90.25	91.68	90.56	92.89	91.78	91.95	93.08	93.98	92.35

Table 12. Accuracy analysis of the BHHO feature selection scheme with the proposed set of LLC dimensionality reduction, clustering, feature extraction and classification models.

	RF	GNB	LR	XGB	NBC	KNN	DT	SVM (Poly)	SVM (RBF)	LGBA
LLC + SC	96.23	95.09	95.33	95.21	94.09	92.23	93.09	95.21	97.31	97.19
LLC + Kmeans	95.44	91.99	93.46	93.22	93.88	92.42	93.88	96.33	97.46	98.08
LLC + FCM	95.56	93.81	94.78	93.47	92.71	91.55	94.71	94.56	97.08	98.12
LLC + HDBSCAN	94.87	94.23	93.98	94.89	92.56	90.79	92.35	94.87	96.92	93.14

From Figure 2, it is obvious that the highest classification accuracy has been obtained for the BHHO feature selection scheme along with the proposed set of LLC dimensionality reduction, FCM clustering, feature extraction and LGBA classifier. From Figure 3, it is obvious that a high classification accuracy is obtained when the Mahalanobis-based feature selection scheme is utilized with the proposed set of LLC dimensionality reduction, K-means clustering, feature extraction and LGBA classifier. When Figure 4 is examined, a high classification accuracy is obtained when the MI feature selection scheme is utilized with the proposed set of LLC dimensionality reduction, SC based analysis, feature extraction and SVM-RBF classifier. When Figure 5 is analyzed, a high classification accuracy is obtained when the BHHO feature selection scheme is utilized with the proposed set of SM dimensionality reduction, FCM clustering, feature extraction and LGBA classifier.

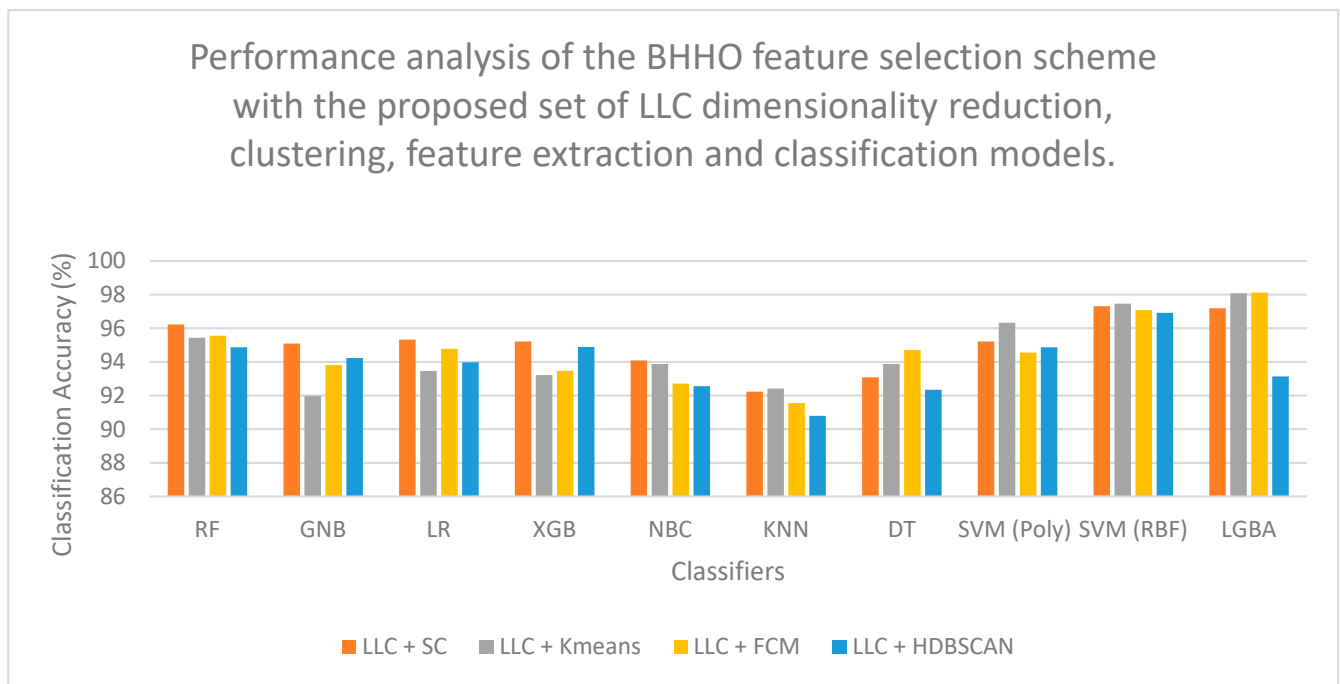


Figure 2. Comparative Analysis of BHHO feature selection scheme with the proposed set of LLC dimensionality reduction, clustering, feature extraction and classification models.

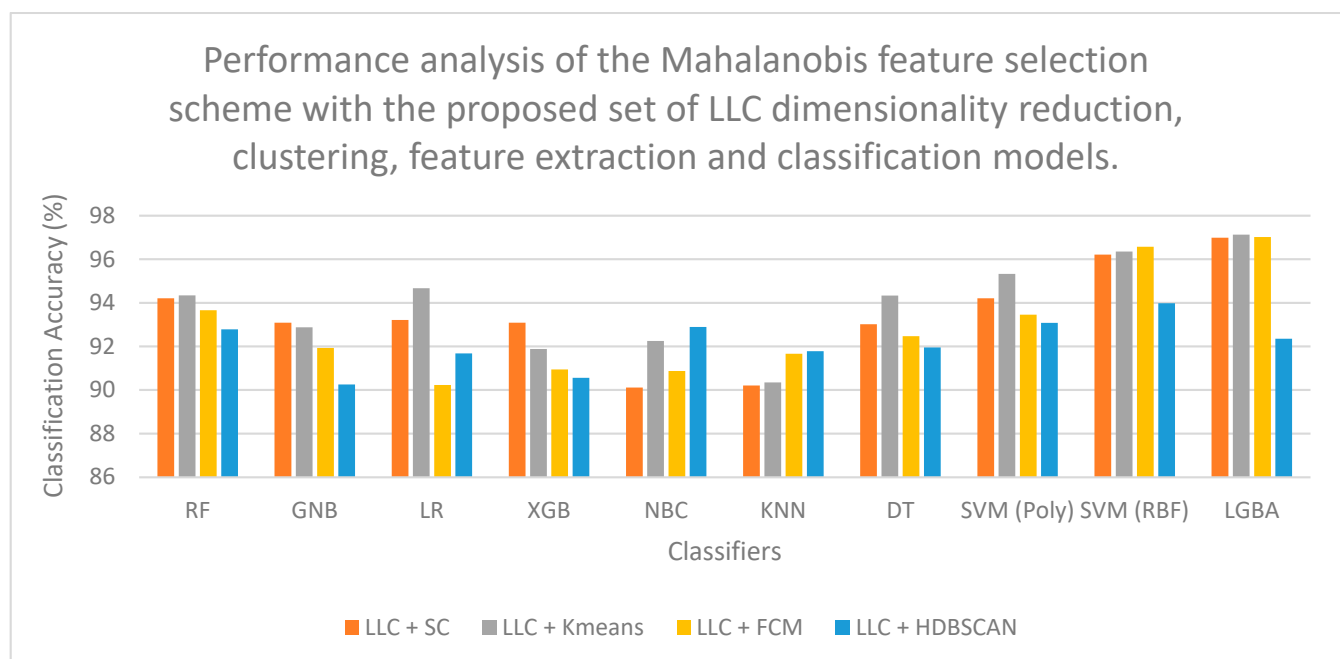


Figure 3. Comparative Analysis of Mahalanobis feature selection scheme with the proposed set of LLC dimensionality reduction, clustering, feature extraction and classification models.

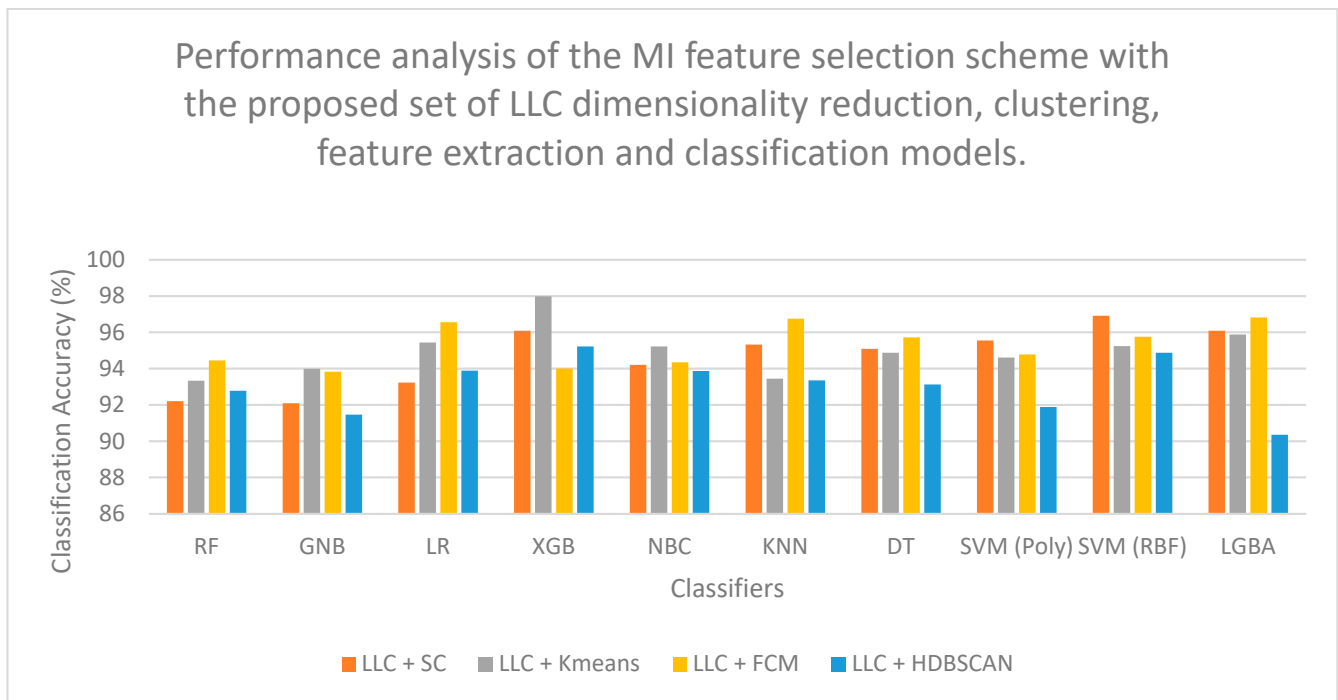


Figure 4. Comparative Analysis of MI feature selection scheme with the proposed set of LLC dimensionality reduction, clustering, feature extraction and classification models.

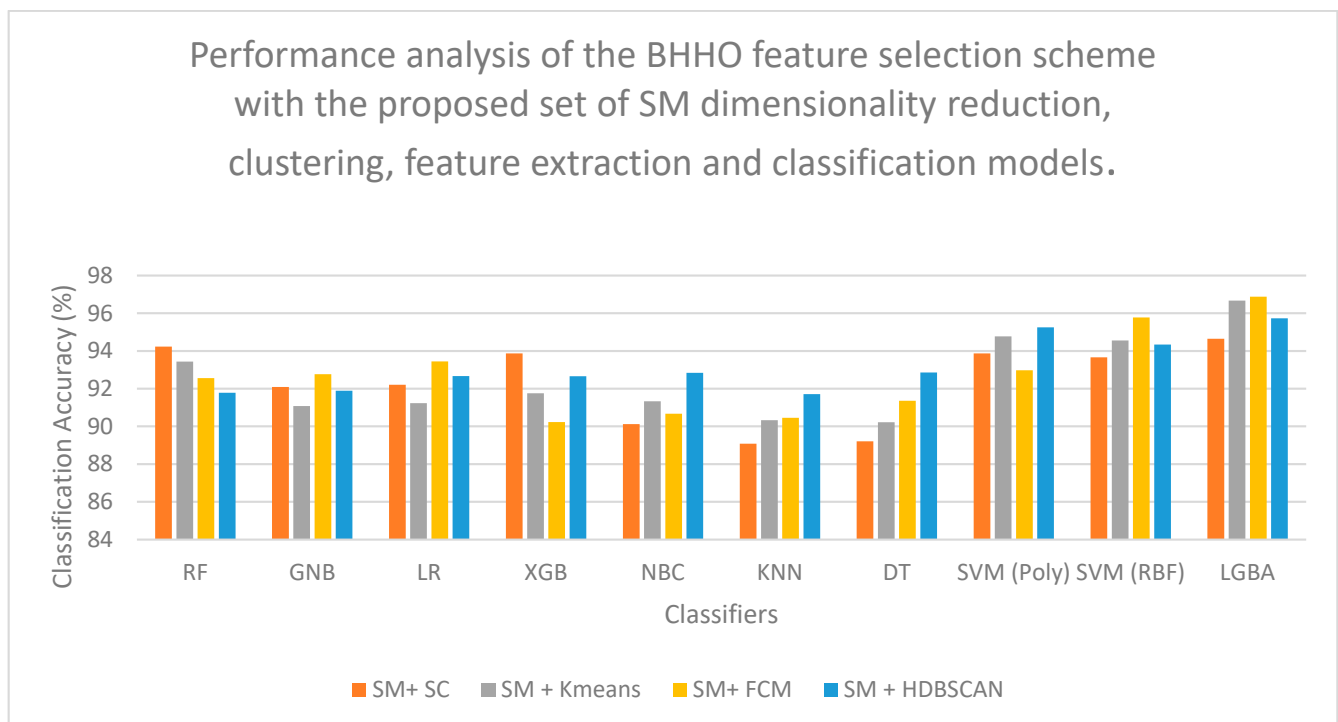


Figure 5. Comparative Analysis of the BHHO feature selection scheme with the proposed set of SM dimensionality reduction, clustering, feature extraction and classification models.

6. Discussion

In this section, the comparison with previous works is made so that a comprehensive understanding of the proposed work can be analyzed well. Moreover, the statistical analysis, computational time and computational complexity are calculated for the proposed models.

6.1. Comparison with Previous Works

The results of the current work are compared with some of the previous work and tabulated in Table 13 as follows.

Table 13. Performance Comparison of the current work with the previous works.

Reference and Year	Method Adopted	Dataset Used	Classification Accuracy (%)
[10], 2008	Concept of Blinded and multi-center validation of EEG for analysis of ADHD	97 ADHD and 36 HC	89
[11], 2011	Decision support system using PSD based analysis	7 ADHD and 3 HC	97
[12], 2012	Quantitative EEG theta/beta ratio analysis with Absolute power analysis	62 ADHD and 36 HC	85
[13], 2016	Nonlinear features with neural network analysis	30 ADHD and 30 HC	93.65
[14], 2016	Attention continuity concept as a nonlinear feature of EEG	29 ADHD and 20 HC	96.7
[15], 2017	Functional brain dynamic analysis with nonlinear features	12 ADHD and 12 HC	83.33
[16], 2019	Nonlinear analysis with machine learning concepts	50 ADHD and 26 HC	96.05
[17], 2019	Transfer learning with CNN concept	50 ADHD and 58 HC	94.39
[18], 2020	Frequency band separation analysis with CNN analysis	31 ADHD and 30 HC	98.48
[19], 2021	Connectivity matrices with directed phase transfer entropy analysis of EEG signals	61 ADHD and 60 HC	89.1
[20], 2022	VMD analysis with Hilbert transform based concepts	61 ADHD and 60 HC	99.95
[21], 2022	Linear and nonlinear effective connectivity patterns	61 ADHD and 60 HC	99
[22], 2022	An eight-pointed star pattern learning network	61 ADHD and 60 HC	89.1
[23], 2022	Dynamic connectivity patterns with ConvLSTM and attention framework concepts	46 ADHD and 45 HC	99.75
[24], 2022	TQWT based analysis with machine learning classifiers	61 ADHD and 60 HC	95.57
[26], 2023	Explainable and Interpretable model using VMD and Hilbert transform analysis	61 ADHD and 60 HC	99.81
[30], 2024	Statistical feature analysis based on Gabor filter	61 ADHD and 60 HC	96.05
[37], 2025	Entropy difference-based EEG selection technique with machine learning classifiers	61 ADHD and 60 HC	99.29
Proposed Works	LLC dimensionality reduction + FCM clustering + IWT feature extraction + BHHO feature selection + LGBA classification	61 ADHD and 60 HC	98.12
	LLC dimensionality reduction + K-means clustering + IWT feature extraction + Mahalanobis analysis-based feature selection + LGBA classification	61 ADHD and 60 HC	97.13
	LLC dimensionality reduction + spectral clustering + IWT feature extraction + MI-based feature selection + SVM-RBF classification	61 ADHD and 60 HC	96.92
	SM dimensionality reduction + FCM clustering + IWT feature extraction + BHHO based feature selection + LGBA classification	61 ADHD and 60 HC	96.88

On analyzing Table 13, it is evident that the proposed works give a good classification accuracy when compared to some of the previous works. Though some of the previous works have obtained slightly higher classification accuracy than the current works, it is to be observed that the proposed unified framework aims to concentrate on hybridizing various techniques together to check its performance and analysis on the ADHD dataset. Various factors, mainly the properties of every unique algorithm, can contribute to the analysis of

results, and so in the future the proposed unified framework can be implemented to other biosignal processing datasets as well.

6.2. Statistical Analysis, Computational Complexity and Computational Time Analysis

A high confidence level was obtained for almost all the features when the standard 2-sided Wilcoxon test was implemented. A good agreement case was reached by the features when the Cohen's Kappa coefficient analysis was tested on them. The Friedman test was implemented on the features and a good variation was found among them, making it fit for classification. The proposed coherent model gave an overall computational complexity of $O(n^3 \log n)$ in this work. The computational time is calculated and shown in Table 14 as follows:

Table 14. Computational Time Analysis for the proposed methods.

Year	Authors	Methods	Computational Time
2026	Proposed works	LLC dimensionality reduction + FCM clustering + IWT feature extraction + BHHO feature selection + LGBA classification	5.012 s
		LLC dimensionality reduction + K-means clustering + IWT feature extraction + Mahalanobis analysis-based feature selection + LGBA classification	6.198 s
		LLC dimensionality reduction + spectral clustering + IWT feature extraction + MI-based feature selection + SVM-RBF classification	8.011 s
		SM dimensionality reduction + FCM clustering + IWT feature extraction + BHHO based feature selection + LGBA classification	6.238 s

6.3. Limitations of the Work

When utilizing the concept of dimensionality reduction, there is a loss of interpretability accompanied by information loss. Also, sometimes it is a huge challenge to choose the optimal components to assess the total number of dimensions in the data and it is accompanied by a lot of linearity assumption constraints, too. When clustering is utilized, it is assumed that the total number of clusters can be subjective at times and is highly sensitive to outliers. Moreover, when clustering is concerned, there are a few shape and size limitations too, and some clustering algorithms like K-means are highly sensitive to the initialization process, depending on the centroid selection stage. When the feature extraction process is concerned, there is a potential for overfitting, and a lot of domain adaptation challenges are present in this context. When the feature selection process is concerned, sometimes a high correlation can be present between variables, which might lead to instability and some feature selection techniques would need a high computational cost as well. When machine learning is concerned, the issue of data dependency, bias concerns, susceptibility to error concerns, and lack of generalization concerns are common. Each module has its own set of merits and drawbacks, but the main beauty of this flow of research is that, when multiple techniques are hybridized together, the intrinsic property of every algorithm works so beautifully to give the best results on a given dataset.

7. Conclusions and Future Works

As there are no definite objective and clinical methods to diagnose ADHD, with the help of EEG signals and machine learning, the detection and classification of ADHD can be done to a certain extent. In this work, a versatile framework is proposed for the classification of ADHD with the help of dimensionality reduction techniques, feature extraction, multiple feature selection techniques and machine learning classifiers. The best

results are obtained when the LLC dimensionality reduction is utilized with FCM clustering and IWT feature extraction, BHHO feature selection and classified with LGBA classifier, reporting a high classification accuracy of 98.12%. The second-best classification accuracy of 97.13% is obtained when the LLC dimensionality reduction is used with K-means clustering, IWT feature extraction, Mahalanobis analysis-based feature selection and LGBA classification. The third-best classification accuracy of 96.92% is obtained when the LLC dimensionality reduction is used with spectral clustering, IWT feature extraction, MI-based feature selection and SVM-RBF classification. Future works aim to work with a variety of other dimensionality reduction techniques and more efficient metaheuristic techniques for feature selection and to test their analysis with machine learning classifiers so that a much higher classification accuracy can be achieved. Also, in the future, more modifications are planned to be made to the proposed feature selection schemes so that their versatility in the classification analysis can be largely improved. In the future, more advanced methodologies such as explainable artificial intelligence are planned to be incorporated along with this model so that it can be used for telemedicine and remote healthcare monitoring services.

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Data Availability Statement: The dataset can be accessed from <https://ieee-dataport.org/open-access/eeg-data-adhd-control-children>. For further details, kindly refer to reference [61,62].

Conflicts of Interest: The authors declare no competing interests.

References

1. Bautista, M.Á.; Hernández-Vela, A.; Escalera, S.; Igual, L.; Pujol, O.; Moya, J.; Violant, V.; Anguera, M.T. A Gesture Recognition System for Detecting Behavioral Patterns of ADHD. *IEEE Trans. Cybern.* **2016**, *46*, 136–147. [\[CrossRef\]](#)
2. Li, G.; Cao, C.; Fu, H.; Li, X.; Gao, X. Modeling Functional Brain Networks for ADHD via Spatial Preservation-Based Neural Architecture Search. *IEEE J. Biomed. Health Inform.* **2024**, *28*, 6854–6864. [\[CrossRef\]](#)
3. Berrezueta-Guzman, J.; Robles-Bykbaev, V.E.; Pau, I.; Pesántez-Avilés, F.; Martín-Ruiz, M.-L. Robotic Technologies in ADHD Care: Literature Review. *IEEE Access* **2022**, *10*, 608–625. [\[CrossRef\]](#)
4. Mandal, S.; Kumar, G.P.; Saini, M.; Satija, U.; Kumar, Y. A Unified Deep Learning Framework for Smartphone-Enabled ADHD Detection. *IEEE Trans. Instrum. Meas.* **2024**, *73*, 6501111. [\[CrossRef\]](#)
5. Ain, Q.U.; Jawed, S.; Subhani, A.R.; Butt, W.H.; Akram, M.U. Examining AI-Powered ADHD Diagnosis: Current Trends, Key Challenges, and Future Directions in the Field. *IEEE Access* **2025**, *13*, 93148–93177. [\[CrossRef\]](#)
6. Oh, S.H.; Park, J.W.; Cho, S.-J. Effectiveness of the VR Cognitive Training for Symptom Relief in Patients with ADHD. *J. Web Eng.* **2022**, *21*, 767–788. [\[CrossRef\]](#)
7. Nash, C.; Nair, R.; Naqvi, S.M. Machine Learning in ADHD and Depression Mental Health Diagnosis: A Survey. *IEEE Access* **2023**, *11*, 86297–86317. [\[CrossRef\]](#)
8. Cao, C.; Li, X.; Wang, M.; Chen, X.; Gao, X. Function-Structural Interaction with Progressive and Multi-Level Feature Fusion for ADHD Classification. *IEEE J. Biomed. Health Inform.* **2025**, *29*, 7568–7578. [\[CrossRef\]](#) [\[PubMed\]](#)
9. Deshpande, G.; Wang, P.; Rangaprakash, D.; Wilamowski, B. Fully Connected Cascade Artificial Neural Network Architecture for Attention Deficit Hyperactivity Disorder Classification from Functional Magnetic Resonance Imaging Data. *IEEE Trans. Cybern.* **2015**, *45*, 2668–2679. [\[CrossRef\]](#)
10. Snyder, S.M.; Quintana, H.; Sexson, S.B.; Knott, P.; Haque, A.; Reynolds, D.A. Blinded, multi-center validation of EEG and rating scales in identifying ADHD within a clinical sample. *Psychiatry Res.* **2008**, *159*, 346–358. [\[CrossRef\]](#)
11. Abibullaev, B.; An, J. Decision support algorithm for diagnosis of ADHD using electroencephalograms. *J. Med. Syst.* **2011**, *36*, 2675–2688. [\[CrossRef\]](#) [\[PubMed\]](#)

12. Ogrim, G.; Kropotov, J.; Hestad, K. The quantitative EEG theta/beta ratio in attention deficit/hyperactivity disorder and normal controls: Sensitivity, specificity, and behavioral correlates. *Psychiatry Res.* **2012**, *198*, 482–488. [\[CrossRef\]](#)
13. Mohammadi, M.R.; Khaleghi, A.; Motie Nasrabadi, A.; Rafiei, S.; Begol, M.; Zarafshan, H. EEG classification of ADHD and normal children using non-linear features and neural network. *Biomed. Eng. Lett.* **2016**, *6*, 66–73.
14. Allahverdy, A.; Khorrami, A.; Mohammadi, M.; Motie Nasrabadi, A. Detecting ADHD children using the attention continuity as nonlinear feature of EEG. *Front. Biomed. Technol.* **2016**, *3*, 28–33.
15. Khoshnoud, S.; Nazari, M.A.; Shamsi, M. Functional brain dynamic analysis of ADHD and control children using nonlinear dynamical features of EEG signals. *J. Integr. Neurosci.* **2018**, *17*, 17–30. [\[CrossRef\]](#)
16. Boroujeni, Y.K. Diagnosis of attention deficit hyperactivity disorder using non-linear analysis of the EEG signal. *IET Syst. Biol.* **2019**, *13*, 260–266. [\[CrossRef\]](#)
17. Zhang, Z.; Li, X. Use transfer learning to promote identification ADHD children with EEG recordings. In Proceedings of the 2019 Chinese Automation Congress, CAC (2019), Hangzhou, China, 22–24 November 2019; pp. 2809–2813.
18. Moghaddari, M.; Lighvan, M.Z.; Danishvar, S. Diagnose ADHD disorder in children using convolutional neural network based on continuous mental task EEG. *Comput. Methods Programs Biomed.* **2020**, *197*, 105738. [\[CrossRef\]](#)
19. Ekhlesi, A.; Motie, N.A.; Mohammadi, M.R. Classification of the children with ADHD and healthy children based on the directed phase transfer entropy of EEG signals. *Front. Biomed. Technol.* **2021**, *8*, 115–122. [\[CrossRef\]](#)
20. Khare, S.K.; Gaikwad, N.B.; Bajaj, V. VHRS: A novel variational mode decomposition and Hilbert transform-based EEG rhythm separation for automatic ADHD detection. *IEEE Trans. Instrum. Meas.* **2022**, *71*, 4008310. [\[CrossRef\]](#)
21. Talebi, N.; Nasrabadi, A.M. Investigating the discrimination of linear and nonlinear effective connectivity patterns of EEG signals in children with attention-deficit/hyperactivity disorder and typically developing children. *Comput. Biol. Med.* **2022**, *148*, 105791. [\[CrossRef\]](#) [\[PubMed\]](#)
22. Tanko, D.; Barua, P.D.; Dogan, S.; Tuncer, T.; Palmer, E.; Ciaccio, E.J.; Acharya, U.R. EPSPatNet86: Eight-pointed star pattern learning network for detection ADHD disorder using EEG signals. *Physiol. Meas.* **2022**, *43*, 035002. [\[CrossRef\]](#)
23. Bakhtyari, M.; Mirzaei, S. ADHD detection using dynamic connectivity patterns of EEG data and ConvLSTM with attention framework. *Biomed. Signal Process. Control* **2022**, *76*, 103708. [\[CrossRef\]](#)
24. Barua, P.; Dogan, S.; Baygin, M.; Tuncer, T.; Palmer, E.; Ciaccio, E.; Acharya, U.R. TMP19: A novel ternary motif pattern-based ADHD detection model using EEG signals. *Diagnostics* **2022**, *12*, 2544. [\[CrossRef\]](#)
25. Joy, R.C.; George, S.T.; Rajan, A.A.; Subathra, M.S.; Sairamya, N.J.; Prasanna, J.; Mohammed, M.A.; Al-Waisy, A.S.; Jaber, M.M.; Al-Andoli, M.N. Detection and classification of ADHD from EEG signals using Tunable Q-Factor Wavelet transform. *J. Sens.* **2022**. [\[CrossRef\]](#)
26. Khare, S.K.; Acharya, U.R. An explainable and interpretable model for attention deficit hyperactivity disorder in children using EEG signals. *Comput. Biol. Med.* **2023**, *155*, 106676. [\[CrossRef\]](#)
27. Esas, M.Y.; Latifoğlu, F. Detection of ADHD from EEG signals using new hybrid decomposition and deep learning techniques. *J. Neural Eng.* **2023**, *20*, 036028. [\[CrossRef\]](#)
28. Alim, A.; Imtiaz, M.H. Automatic Identification of Children with ADHD from EEG Brain Waves. *Signals* **2023**, *4*, 193–205. [\[CrossRef\]](#)
29. Maniruzzaman, M.; Hasan, M.A.M.; Asai, N.; Shin, J. Optimal Channels and Features Selection Based ADHD Detection from EEG Signal Using Statistical and Machine Learning Techniques. *IEEE Access* **2023**, *11*, 33570–33583. [\[CrossRef\]](#)
30. Sathiya, E.; Rao, T.D.; Kumar, T.S. Gabor filter-based statistical features for ADHD detection. *Front Hum Neurosci.* **2024**, *18*, 1369862. [\[CrossRef\]](#) [\[PubMed\]](#)
31. Cura, O.K.; Akan, A.; Atli, S.K. Detection of Attention deficit hyperactivity disorder based on EEG feature maps and deep learning. *Biocybern. Biomed. Eng.* **2024**, *44*, 450–460. [\[CrossRef\]](#)
32. Loh, H.W.; Ooi, C.P.; Oh, S.L.; Barua, P.D.; Tan, Y.R.; Acharya, U.R.; Fung, D.S. ADHD/CD-NET: Automated EEG-based characterization of ADHD and CD using explainable deep neural network technique. *Cogn. Neurodyn.* **2024**, *18*, 1609–1625. [\[CrossRef\]](#) [\[PubMed\]](#)
33. Attallah, O. ADHD-AID: Aiding Tool for Detecting Children's Attention Deficit Hyperactivity Disorder via EEG-Based Multi-Resolution Analysis and Feature Selection. *Biomimetics* **2024**, *9*, 188. [\[CrossRef\]](#) [\[PubMed\]](#)
34. Latifi, B.; Amini, A.; Nasrabadi, A.M. Siamese based deep neural network for ADHD detection using EEG signals. *Comput. Biol. Med.* **2024**, *182*, 109092. [\[CrossRef\]](#) [\[PubMed\]](#)
35. Abedian, S. Diagnosis of Adult ADHD Using EEG Signals Based on the Spectrogram and Convolutional Neural Networks. *Int. J. Comput. Intell. Appl.* **2024**, *23*, 2350034. [\[CrossRef\]](#)
36. García-Ponsoda, S.; Maté, A.; Trujillo, J. Refining ADHD diagnosis with EEG: The impact of preprocessing and temporal segmentation on classification accuracy. *Comput. Biol. Med.* **2024**, *183*, 109305. [\[CrossRef\]](#)
37. Maheshwari, S.; Rajesh, K.N.V.P.S.; Kanhangad, V.; Rajendra Acharya, U.; Sunil Kumar, T. Entropy difference-based EEG channel selection technique for automated detection of ADHD. *PLoS ONE* **2025**, *20*, e0319487. [\[CrossRef\]](#)

38. Li, L.; Guo, X.; Yang, Z.; Zhao, Y.; Liu, X.; Yang, J.; Chen, Y.; Peng, X.; Han, L. ADHD detection from EEG signals using GCN based on multi-domain features. *Front. Neurosci.* **2025**, *19*, 1561994.
39. Alhussen, A.; Alutaibi, A.I.; Sharma, S.K.; Khan, A.R.; Ahmad, F.; Tejani, G.G. Early attention-deficit/hyperactivity disorder (ADHD) with NeuroDCT-ICA and rhinofish optimization (RFO) algorithm based optimized ADHD-AttentionNet. *Sci. Rep.* **2025**, *15*, 6967. [\[CrossRef\]](#)
40. Mao, Y.; Qi, X.; He, L.; Wang, S.; Wang, Z.; Wang, F. Advanced machine learning techniques reveal multidimensional EEG abnormalities in children with ADHD: A framework for automatic diagnosis. *Front. Psychiatry* **2025**, *16*, 1475936. [\[CrossRef\]](#)
41. Bansal, J.; Gangwar, G.; Aljaidi, M.; Alkoradees, A.; Singh, G. EEG-Based ADHD Classification Using Autoencoder Feature Extraction and ResNet with Double Augmented Attention Mechanism. *Brain Sci.* **2025**, *15*, 95. [\[CrossRef\]](#)
42. Herwanto, H.W.; Handayani, A.N.; Wibawa, A.P.; Chandrika, K.L.; Arai, K. Comparison of Min-Max, Z-Score and Decimal Scaling Normalization for Zoning Feature Extraction on Javanese Character Recognition. In Proceedings of the 2021 7th International Conference on Electrical, Electronics and Information Engineering (ICEEIE), Malang, Indonesia, 2 October 2021; pp. 1–3.
43. Li, R.; Liu, L. Research on rolling bearing fault diagnosis based on improved local linear embedding algorithm. In Proceedings of the 2022 Global Reliability and Prognostics and Health Management (PHM-Yantai), Yantai, China, 13–16 October 2022; pp. 1–3.
44. Pohlheim, H. Multidimensional Scaling for Evolutionary Algorithms—Visualization of the Path through Search Space and Solution Space Using Sammon Mapping. *Artif. Life* **2006**, *12*, 203–209. [\[CrossRef\]](#)
45. Huang, Q.; Wang, H.; Xu, Q.; Bi, W. Semi-supervised Learning with Locally Linear Coordination for Face Recognition. In Proceedings of the 2009 Fifth International Conference on Natural Computation, Tianjian, China, 14–16 August 2009; pp. 255–259.
46. Huang, D.; Wang, C.-D.; Wu, J.-S.; Lai, J.-H.; Kwok, C.-K. Ultra-Scalable Spectral Clustering and Ensemble Clustering. *IEEE Trans. Knowl. Data Eng.* **2020**, *32*, 1212–1226. [\[CrossRef\]](#)
47. Chen, T.S.; Tsai, T.H.; Chen, Y.T.; Lin, C.C.; Chen, R.C.; Li, S.Y.; Chen, H.Y. A combined K-means and hierarchical clustering method for improving the clustering efficiency of microarray. In Proceedings of the 2005 International Symposium on Intelligent Signal Processing and Communication Systems, Hong Kong, China, 13–16 December 2005; pp. 405–408.
48. Kmita, K.; Kaczmarek-Majer, K.; Hryniewicz, O. Explainable Impact of Partial Supervision in Semi-Supervised Fuzzy Clustering. *IEEE Trans. Fuzzy Syst.* **2024**, *32*, 3189–3198. [\[CrossRef\]](#)
49. He, S.; Wu, C. A Fréchet Distance-Hierarchical Density-based Spatial Clustering of Applications with Noise for Ship Trajectory Clustering and Route Recognition. In Proceedings of the 2024 International Conference on Distributed Systems, Computer Networks and Cybersecurity (ICDSCNC), Bengaluru, India, 20–21 September 2024; pp. 1–5.
50. Guo, J.-H.; Lin, J.-X.; Zhang, Z.-C.; Ling, H.-Y. CDBSCAN: Density clustering based on silhouette coefficient constraints. In Proceedings of the 2022 International Conference on Computer Engineering and Artificial Intelligence (ICCEAI), Shijiazhuang, China, 22–24 July 2022; pp. 600–605.
51. Chaudhury, K.N.; Unser, M. Construction of Hilbert Transform Pairs of Wavelet Bases and Gabor-Like Transforms. *IEEE Trans. Signal Process.* **2009**, *57*, 3411–3425. [\[CrossRef\]](#)
52. Inoue, K.; Takajo, A.; Maeda, M.; Matsuoka, S. Tuning method of modified wavelet transform in human sleep EEG analysis. In Proceedings of the 2007 International Conference on Control, Automation and Systems, Seoul, Republic of Korea, 17–20 October 2007; pp. 2784–2789.
53. Nielsen, F.; Nock, R. On the chi square and higher-order chi distances for approximating f-divergences. *IEEE Signal Process. Lett.* **2014**, *21*, 10–13. [\[CrossRef\]](#)
54. Jakobsen, S.K. Mutual Information Matrices Are Not Always Positive Semidefinite. *IEEE Trans. Inf. Theory* **2014**, *60*, 2694–2696. [\[CrossRef\]](#)
55. Washizawa, Y.; Hotta, S. Mahalanobis Distance on Extended Grassmann Manifolds for Variational Pattern Analysis. *IEEE Trans. Neural Netw. Learn. Syst.* **2014**, *25*, 1980–1990. [\[CrossRef\]](#)
56. MiarNaeimi, F.; Azizyan, G.; Rashki, M. Horse herd optimization algorithm: A nature-inspired algorithm for high-dimensional optimization problems. *Knowl. Based Syst.* **2021**, *213*, 106711. [\[CrossRef\]](#)
57. Elmanakhly, D.A.; Saleh, M.; Rashed, E.A.; Abdel-Basset, M. BinHOA: Efficient Binary Horse Herd Optimization Method for Feature Selection: Analysis and Validations. *IEEE Access* **2022**, *10*, 26795–26816. [\[CrossRef\]](#)
58. Pandey, L.B.; Sharma, M.; Tiwari, R.; Panda, R.S.; Roy, P. A Survey of Optimized Compiler Using Advanced Machine learning and Deep Learning Techniques. In Proceedings of the 2024 IEEE 6th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA), Hamburg, Germany, 19–20 October 2024; pp. 256–259.
59. Vapnik, V. SVM method of estimating density, conditional probability, and conditional density. In Proceedings of the 2000 IEEE International Symposium on Circuits and Systems (ISCAS), Geneva, Switzerland, 28–31 May 2000; Volume 2, pp. 749–752.
60. Taha, A.A.; Malebary, S.J. An Intelligent Approach to Credit Card Fraud Detection Using an Optimized Light Gradient Boosting Machine. *IEEE Access* **2020**, *8*, 25579–25587. [\[CrossRef\]](#)

61. Nasrabadi, A.; Allahverdy, A.; Samavati, M.; Mohammadi, M. *EEG Data for ADHD/Control Children*; IEEE Dataport: Piscataway, NJ, USA, 2020.
62. Samavati, M.; Motie Nasrabadi, A.; Mohammadi, M.R. Automatic minimization of eye blink artifacts using fractal dimension of independent components of multichannel EEG. In Proceedings of the 20th Iranian Conference on Electrical Engineering (ICEE2012), Tehran, Iran, 15–17 May 2012; pp. 1576–1578.

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